Probabilistic Machine Learning

Lecture 02: Probability and Statistics

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Where We Left Off

Lecture 01 ended on an ideal — model the joint, condition on the data, read off answers — and on the shortcuts most methods take. Today builds the language all of them are written in:

Model the joint

Module 3 — random variables and their joint distribution

→

Condition on the data

Module 2 for events — Module 4 for a parameter

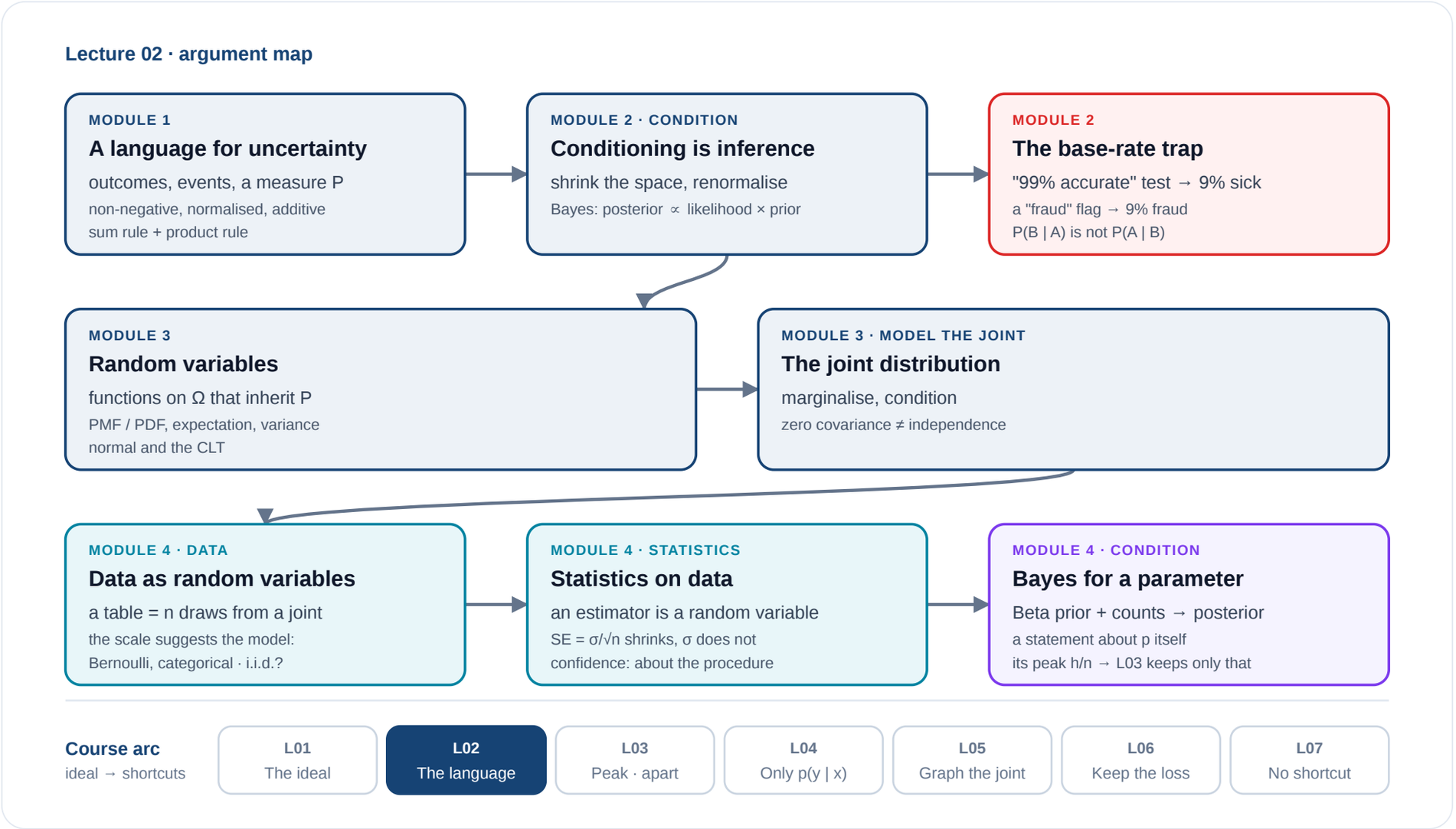
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Read off answers

Module 4 — a peak, a mean, an interval

It also left a claim unpaid: that **epistemic** uncertainty is the kind more data removes. Module 4 turns that slogan into a rate.

Argument Map



Module 1

A Language for Uncertainty

> ***Guiding Question:*** *We need a language that can say "how strongly", not just "true or false". What must such a language satisfy — and who decides the numbers?*

Example: Two Fruit Boxes

Red box
2 apples, 6 oranges

Blue box
3 apples, 1 orange

We pick a box, then a fruit from it.

- If we pick the **red box 40%** of the time, how often do we get an apple?
- We got an **orange** — which box did it probably come from?

The Counts, in One Grid

A hundred picks with the red box chosen 40% of the time, as a table of counts — box across, fruit down:

	$B = r$	$B = b$	row total
$F = a$ (apple)	10	45	**55**
$F = o$ (orange)	30	15	**45**
column total	**40**	**60**	**100**

Joint
a single cell over the total
$P(B = r, F = a) = \frac{10}{100}$

Marginal
a row or column total over the total
$P(F = a) = \frac{55}{100}$

Conditional
a cell over its own row or column
$P(F = a \mid B = r) = \frac{10}{40}$

The Two Rules

Sum rule

$$P(X) = \sum_{Y} P(X, Y)$$

Add along a row or column to remove a variable.

Product rule

$$P(X, Y) = P(Y \mid X) P(X)$$

A cell is its column total times its share of that column.

Almost all of probability theory is these two rules applied repeatedly.

Bayes' Theorem, in Advance

Combining the two rules and the symmetry $P(X, Y) = P(Y, X)$:

$$P(Y \mid X) = \frac{P(X \mid Y)P(Y)}{P(X)}, \qquad P(X) = \sum_{Y} P(X \mid Y)P(Y)$$

Which box did the orange come from?

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Which box did the orange come from?

$$P(B = r \mid F = o) = \frac{P(F = o \mid B = r)P(B = r)}{P(F = o)} = \frac{\frac{6}{8} \cdot 0.4}{\frac{6}{8} \cdot 0.4 + \frac{1}{4} \cdot 0.6} = \frac{2}{3}$$

Before the pick: 40% red. After seeing an orange: 67% red. The orange is **evidence** — because the red box is full of oranges and the blue one is not.

Bayes' Theorem, in Advance

Combining the two rules and the symmetry $P(X, Y) = P(Y, X)$:

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Before the pick: 40% red. After seeing an orange: 67% red. The orange is **evidence** — because the red box is full of oranges and the blue one is not.

And if the blue box held 3 apples and 9 oranges — the same orange share as the red one?

Takeaway

Two rules — **sum** to remove a variable, **product** to split a joint — generate joint, marginal, conditional probabilities and Bayes' theorem. What follows makes them precise and extends them beyond counting.

What Makes an Experiment "Random"?

Definition (Random experiment). A procedure that can — at least in theory — be repeated infinitely often, characterised by:

1. **Configuration** — a precisely specified system that can be reconstructed.
2. **Procedure** — an instruction for executing the experiment.
3. **Unpredictability** of the outcome.

Coin flip: a 1-Euro coin (configuration), placed on the thumb and snapped into the air (procedure), record heads or tails (outcome).

Randomness as Missing Information

> A random experiment is still causal. Its unpredictability reflects **missing information about the causal chain**, not an absence of causes.
>
> Consequence: a random experiment can become a deterministic process once new information is available.

A coin flip filmed at 10,000 fps and measured precisely enough stops being random. What does that imply for a model that calls something "random"?

Sample Space and Events

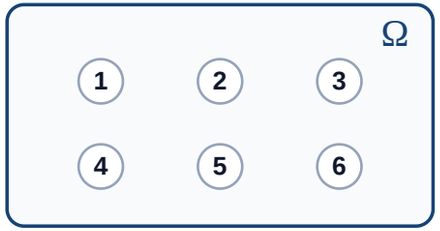
Definition (Sample space, event space). A set $\Omega = \{\omega_1, \omega_2, \ldots, \omega_n\}$ is the **sample space** of a random experiment if every outcome is associated with exactly one element $\omega \in \Omega$. The elements are called **outcomes**.

Let Ω be finite. Every subset $A \subseteq \Omega$ is an **event**; the event A **occurs** iff the observed outcome ω is a member of A. The set of all events, the power set $\mathcal{P}(\Omega)$, is the **event space**.

An outcome is a single result; an event is a *question you can ask* about the result.

For infinite Ω — the drug dosage below — the event space is a suitable family of subsets rather than the full power set; we return to this when continuous distributions appear.

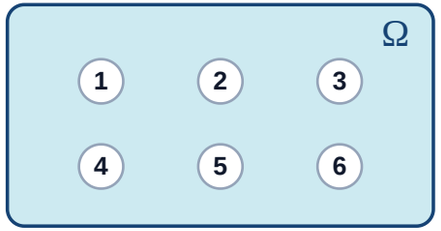
Important Event Types



$\varnothing$

impossible event

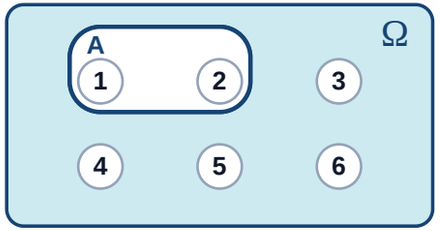
no outcome belongs to it



Ω

certain event

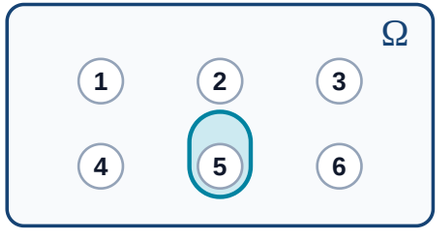
every outcome belongs to it



$\bar{A} = \Omega \setminus A$

complementary event

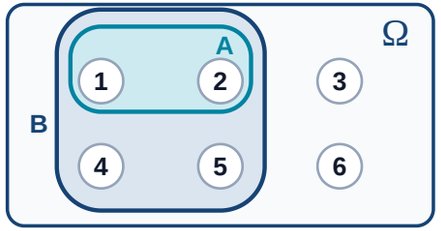
A = {1, 2}, $\bar{A}$ = {3, 4, 5, 6}



$|A| = 1$

elementary event

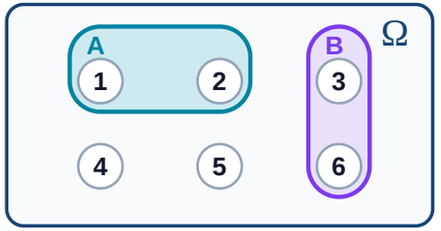
A = {5}: one outcome



$A \subseteq B$

A entails B

A = {1, 2}, B = {1, 2, 4, 5}



$A \cap B = \varnothing$

incompatible events

A = {1, 2}, B = {3, 6}

Examples

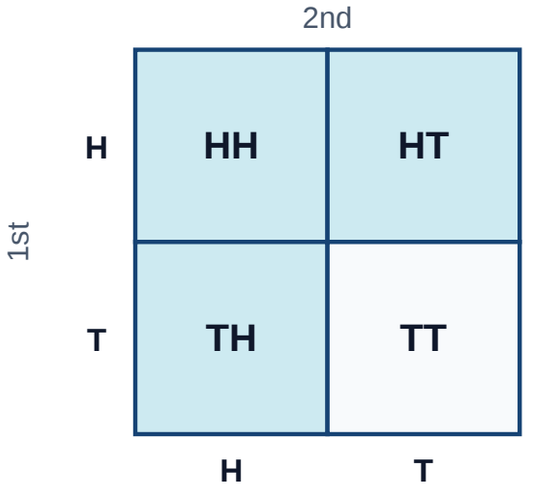
One coin flip



E = {H}: "heads"

all events: ∅, {H}, {T}, {H, T}

Two coin flips



"at least one head"

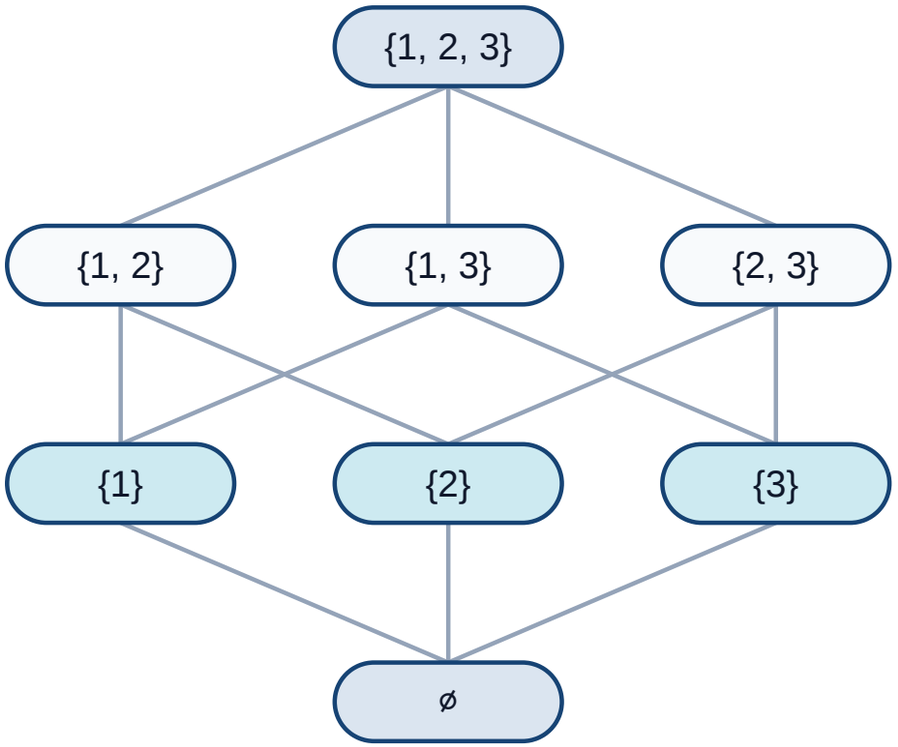
= {HH, HT, TH}

4 outcomes

2^4 = 16 possible events

An event is a set of outcomes you can shade — "heads", "at least one head" — and the sample space fixes which such questions can be asked.

The Event Space Grows Fast



$|\Omega| = n \rightarrow 2^n$ events

1	2
2	4
3	8
6	64
10	1,024
20	1,048,576

lines: one element added

For $|\Omega| = n$ the event space has 2^n elements — far too many to list, which is why probability is done by counting, not by enumeration.

Not Every Sample Space Is Finite

Dosage of a drug: Ω = (0, ∞)



uncountably many outcomes — probability can only live on intervals, not on points

Race with 7 teams: all orderings

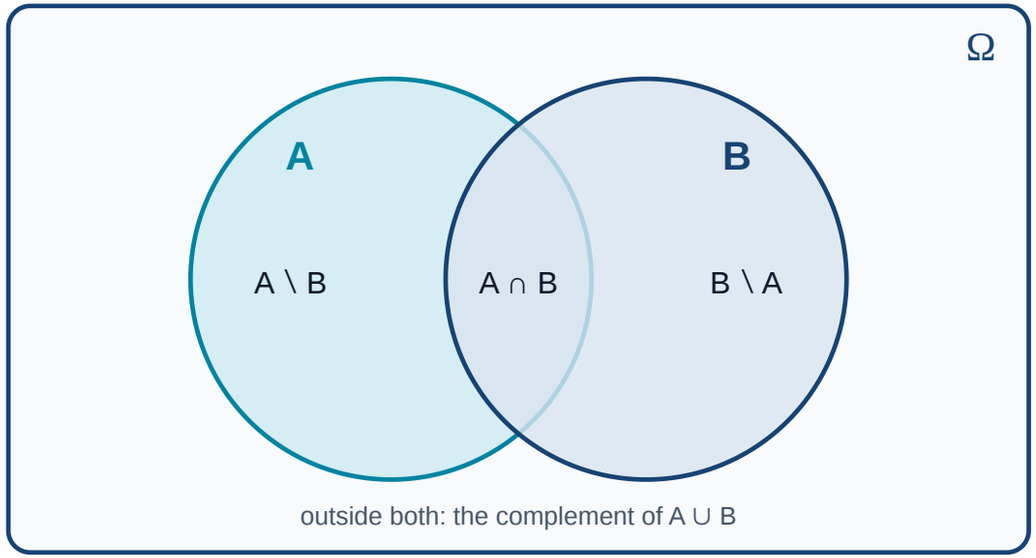


7! = 5,040 outcomes; shaded: the event "team 3 wins"

Choosing Ω is a modelling decision: it fixes what counts as an outcome, and therefore which questions can be asked at all.

Venn Diagrams

Venn diagrams are the graphical representation of events:

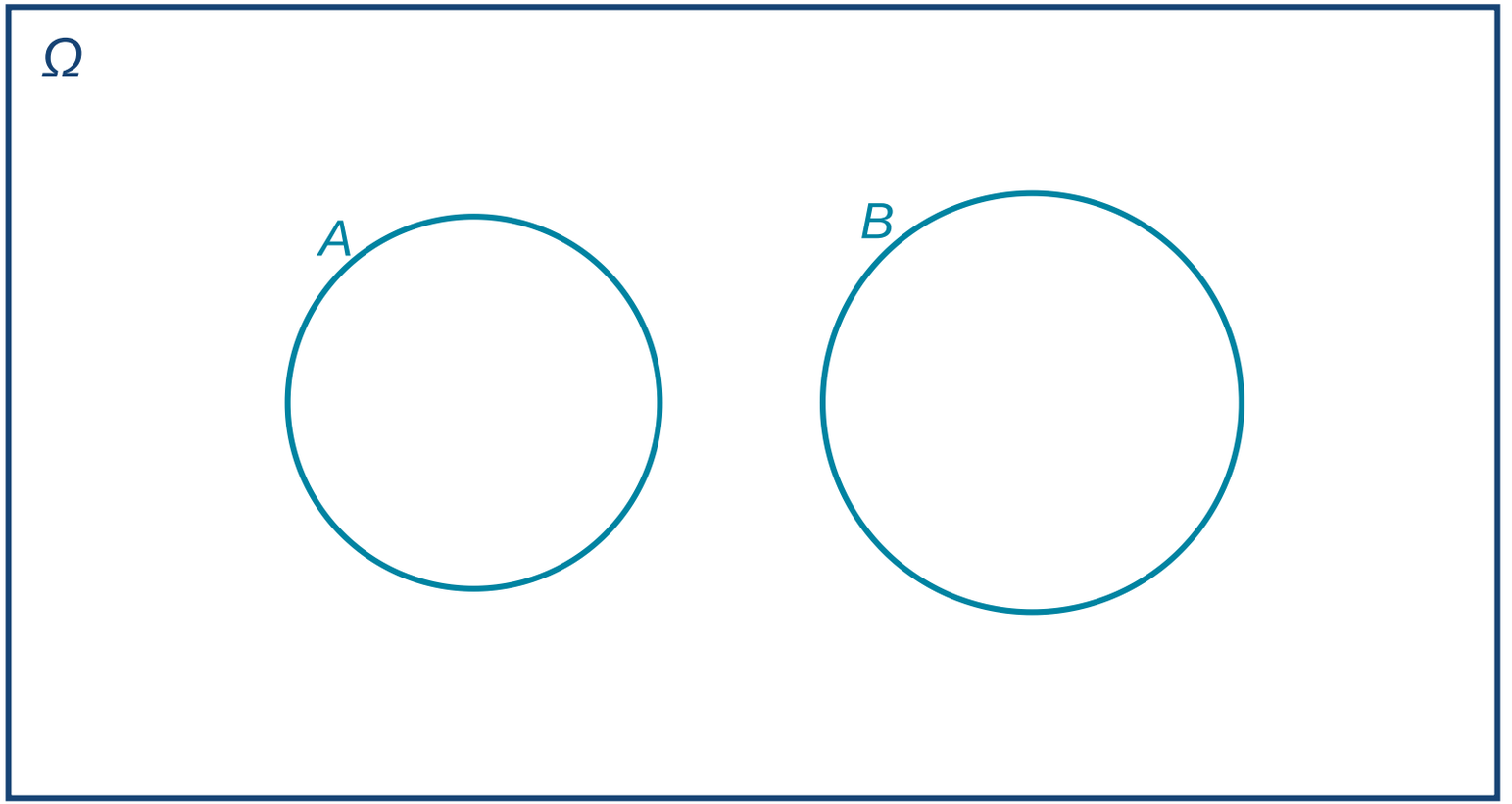


- the **rectangle** is the sample space Ω
- a **circle** is an event $A \subseteq \Omega$
- every **region** is a set — and every set operation is a region you can point at

Important Event Types

Before each step: which region will light up?

$$A \cap B = \emptyset$$



A and B are disjoint

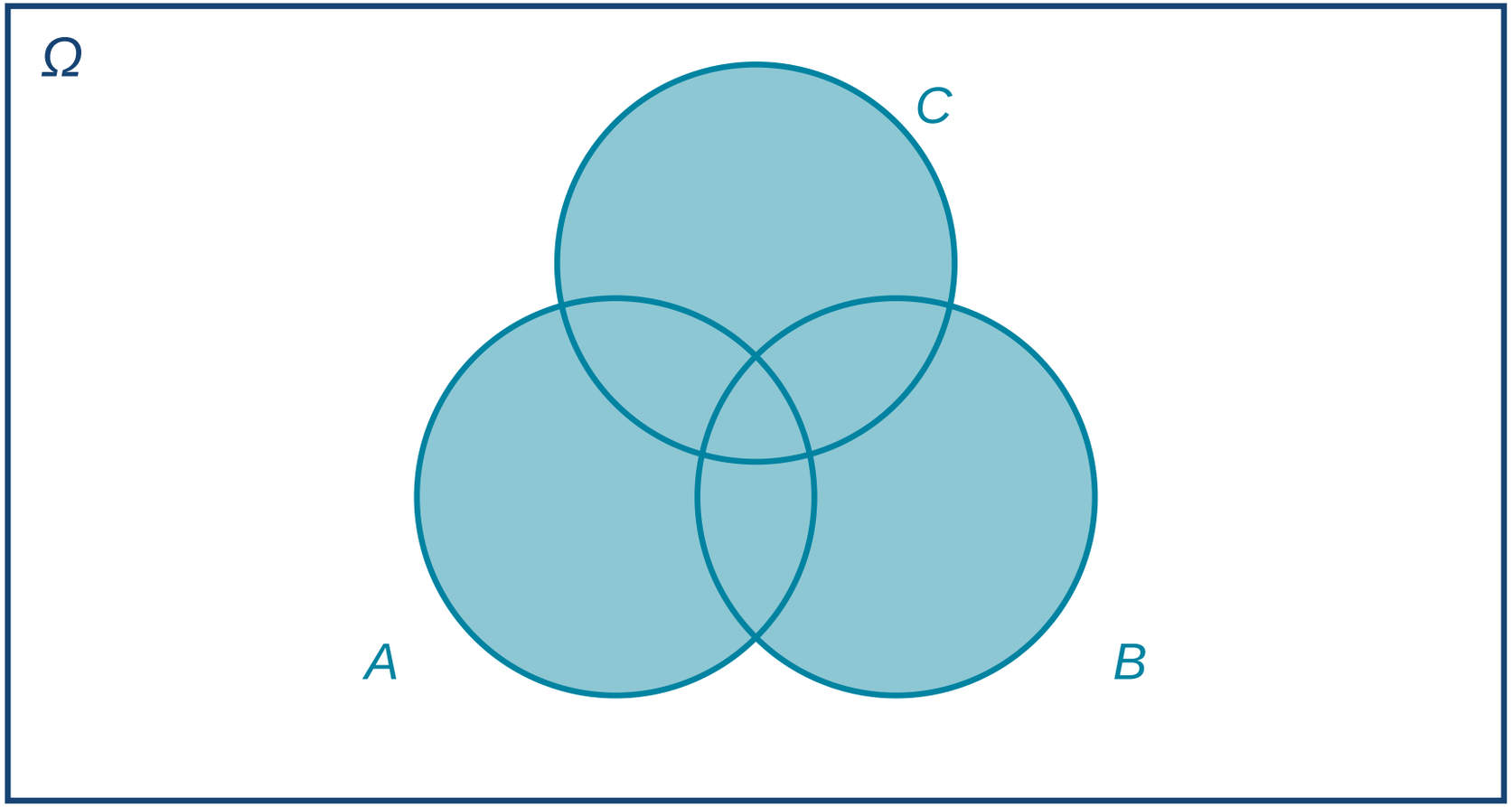
Algebra of Events

Law	Union form	Intersection form
Commutative	$A \cup B = B \cup A$	$A \cap B = B \cap A$
Associative	$(A \cup B) \cup C = A \cup (B \cup C)$	$(A \cap B) \cap C = A \cap (B \cap C)$
Distributive	$(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$	$(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$

Events form a set algebra — the same laws you know, but now every identity is a picture.

Set Operations as Regions

A ∪ B ∪ C



Union of three events

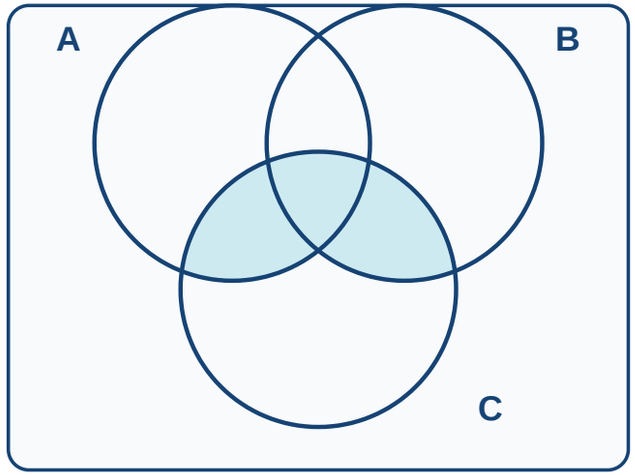
Mini-Exercise

Are $(A \cup B) \cap C$ and $(A \cap B) \cup C$ the same region?

Mini-Exercise

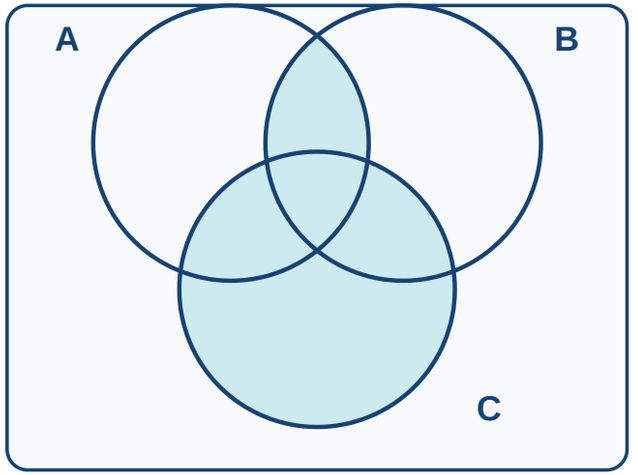
Are $(A \cup B) \cap C$ and $(A \cap B) \cup C$ the same region?

(A ∪ B) ∩ C



inside C only

(A ∩ B) ∪ C



all of C, plus the lens A ∩ B

No. The first lies entirely inside C; the second is all of C plus the lens $A \cap B$. The distributive laws expand them into different expressions.

An Intuition: Probability as Area

Draw a Venn diagram on the ground, with each area proportional to the number of elementary events it contains. Throw a stone into the rectangle at random.

The probability of A is how likely the stone lands inside A, assuming

1. we hit the rectangle Ω for certain, and
2. every part of Ω is equally likely to be hit.

Probability is then a **measure of the state** $A \subseteq \Omega$ of a random experiment — literally, how much of the sample space the event occupies.

> Both assumptions are doing real work. Assumption 2 is the one we will have to give up.

Two Ways to Read a Probability

Frequentist

The probability of an elementary event a is the fraction of repetitions of the experiment that yield a.

Requires the experiment to be repeatable.

Bayesian

The probability of a is the degree of certainty that a will happen — even without conducting the experiment.

Requires a stated state of knowledge.

Both interpretations yield the same mathematics. They are mental models for what the number *means*, not different theories.

The Classical (Laplace) Definition

Definition (Classical / Laplace probability). If every elementary event in Ω is assigned the same probability, then

$$P(A) = \frac{|A|}{|\Omega|} = \frac{\text{number of cases favourable for } A}{\text{number of possible outcomes}}$$

An experiment whose configuration and procedure make outcomes equiprobable — by definition or by construction — is a **Laplace experiment**; its probabilities need not be estimated, they are implied by the setup.

Why That Definition Is Not Enough

> **The circularity.** "Laplace probability" defines probability using *equiprobability* — that is, using another notion of probability. Strictly speaking it is a circular definition.

The frequentist repair — defining probability as the limit of relative frequencies — fails too: such a limit exists only in mathematical settings, never in an experiment that can actually be repeated.

So what do we do? Stop trying to *define* what probability is, and instead specify what it must *do*.

The Kolmogorov Axioms

Definition (Probability measure, Kolmogorov 1933). Let Ω be a sample space and $\mathcal{P}(\Omega)$ its event space. A function $P : \mathcal{P}(\Omega) \to \mathbb{R}$ is a **probability measure** if:

Axiom I

$$P(A) \geq 0$$

non-negative

Axiom II

$$P(\Omega) = 1$$

normalised

Axiom III

$$A \cap B = \emptyset \Rightarrow$$
$$P(A \cup B) = P(A) + P(B)$$

additive

Nothing is said about *which* probabilities the events get — only about how any assignment must behave.

Beyond finite sample spaces. When Ω is infinite, the event space is not the full power set but a **σ-algebra** of measurable sets, and Axiom III is required for **countably many** disjoint events. Everything we compute in this course is unaffected; the construction belongs to measure theory.

Probability Space

Definition. Let Ω be a sample space, $\mathcal{P}(\Omega)$ an event space, and P a probability measure. The triple $(\Omega, \mathcal{P}(\Omega), P)$ — also written (Ω, P) — is a **probability space**.

Every probabilistic statement in this course is made inside some probability space. Naming it is the discipline that keeps modelling assumptions visible.

What the Axioms Already Give Us

$$P(A) + P(\overline{A}) = 1$$

(Axioms II, III)

$$P(\emptyset) = 0$$

(from the above with $A = \Omega$)

$$A \subseteq B \Rightarrow P(A) \leq P(B)$$

monotonicity *(Axioms I, III)*

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

inclusion–exclusion *(Axiom III)*

For mutually exclusive $A_1, \ldots, A_p$: $P(A_1 \cup \ldots \cup A_p) = P(A_1) + \ldots + P(A_p)$.

Takeaway

Probability is not defined by counting or by repetition — it is any assignment of numbers to events that is **non-negative, normalised, and additive**. The Laplace formula is one such assignment, not the definition.

Module 2

Conditioning Is Inference

> ***Guiding Question:*** *A 99%-accurate test says you are sick. Why is that not a 99% chance of being sick — and what exactly does the evidence do to your belief?*

Motivating Example

You roll a die twice. **After the first roll you see a 3.** Knowing this, what is the probability that both rolls sum to 8?

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You roll a die twice. **After the first roll you see a 3.** Knowing this, what is the probability that both rolls sum to 8?

- Before the first roll: $\Omega = \{(1,1),(1,2),\ldots,(6,6)\}$, $|\Omega| = 36$
- After seeing the 3, only $\Omega_{red} = \{(3,1),\ldots,(3,6)\}$ remains
- Outcomes like $(2,4)$ have become **impossible**
- The reduced space now carries all the probability: $P(\Omega_{red}) = 1$
- "Sum is 8" inside it: $E = \{(3,5)\}$

$$P = \frac{1}{6}$$

Definition

Definition (Conditional probability).

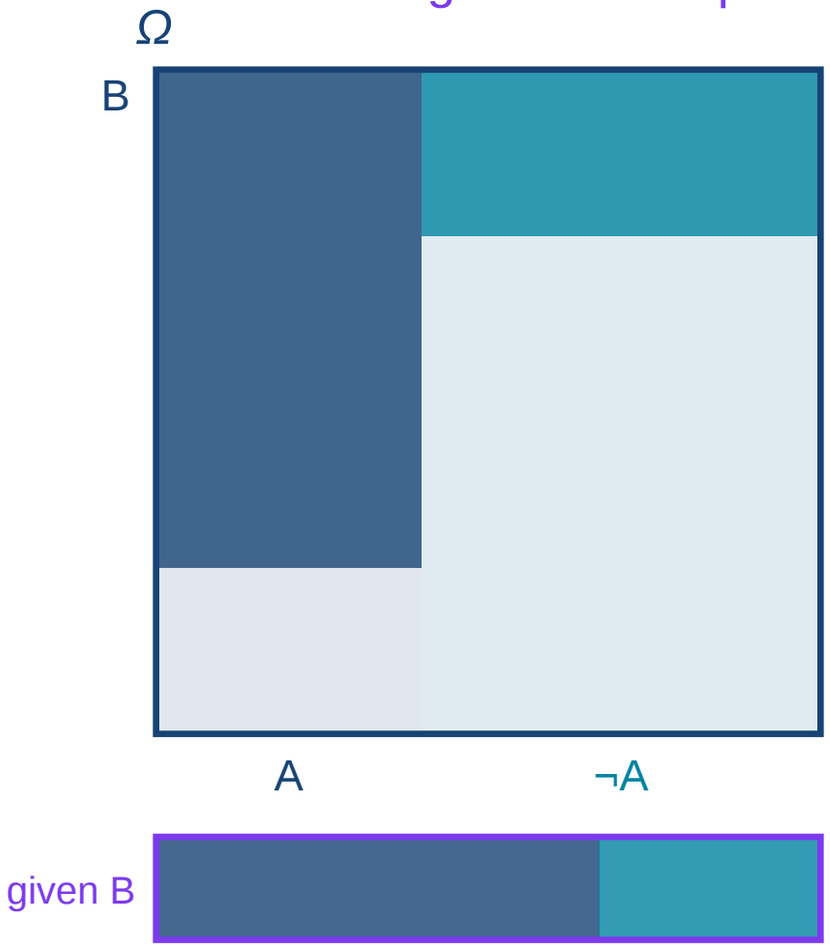
$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad \text{if } P(B) > 0$$

$P(A \mid B)$ is the probability of A **under the condition** B.

Back to the stone: *given that the stone landed in area B, how likely is it that it also landed in A?* We renormalise by the area of B, because B is now the whole world.

Conditioning, Drawn

Conditioning on B = keep the B cells, stretch them back to width 1



P(A) = 0.40

P(B | A) = 0.75, P(B | ¬A) = 0.25

P(B) = 0.45

P(A ∩ B) = 0.30

P(A | B) = 0.67

Before the last click: what must the two shaded cells be divided by?

Example: Two Coins

Two coins are flipped. The event of interest is $B = \{(h, h)\}$.

(a) "The first coin is heads"

$F = \{(h, h), (h, t)\}$

$$P(B \mid F) = \frac{P(B \cap F)}{P(F)} = \frac{1/4}{1/2} = \frac{1}{2}$$

(b) "At least one coin is heads"

$A = \{(h, h), (h, t), (t, h)\}$

$$P(B \mid A) = \frac{P(B \cap A)}{P(A)} = \frac{1/4}{3/4} = \frac{1}{3}$$

Same event B, two different pieces of information, two different answers. **Why is (b) smaller?**

Total Probability

Theorem (Total probability). Let $B_1, \ldots, B_k$ be mutually exclusive with $\Omega = B_1 \cup \ldots \cup B_k$ and $P(B_i) > 0$. Then for any event A:

$$P(A) = \sum_{i=1}^{k} P(B_i) \cdot P(A \mid B_i)$$

Proof. Decompose A along the partition and use additivity:

$$P(A) = P((B_1 \cup \ldots \cup B_k) \cap A) = \sum_{i=1}^{k} P(B_i \cap A) = \sum_{i=1}^{k} P(B_i) \cdot P(A \mid B_i)$$

Rules That Follow

Product rule

$$P(A \cap B) = P(B) \cdot P(A \mid B)$$
$$= P(A) \cdot P(B \mid A)$$

Reversing the condition

$$P(B \mid A) = \frac{P(B) \cdot P(A \mid B)}{P(A)}$$

Complement, same condition

$$P(\overline{A} \mid B) = 1 - P(A \mid B)$$

But in general

$$P(A \mid \overline{B}) \neq 1 - P(A \mid B)$$

Held fixed at B, the map $A \mapsto P(A \mid B)$ satisfies the Kolmogorov axioms — conditioning yields a **new probability measure** P_B.

Chain Rule

$$P(E_1 \cap E_2 \cap \ldots \cap E_n) = P(E_1) \cdot P(E_2 \mid E_1) \cdot P(E_3 \mid E_1 \cap E_2) \cdots P(E_n \mid E_1 \cap \ldots \cap E_{n-1})$$

By successive application of the product rule.

> Any joint probability can be written as a chain of conditionals. Choosing a *good* order — and dropping conditions that do not matter — is exactly what graphical models will do later.

Takeaway

> Conditioning **shrinks the sample space and renormalises**. Everything else — product rule, total probability, Bayes, the chain rule — is bookkeeping on that single move.

Bayes' Theorem — An Intuition First

A test for a rare disease is **99% accurate**: positive for 99% of the sick, negative for 99% of the healthy. About **1 in 1,000 people** has the disease.

Your test comes back positive. How likely is it that you are actually sick?

Bayes' Theorem — An Intuition First

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Your test comes back positive. How likely is it that you are actually sick?

> Only about **9%** — the answer most people miss by a factor of ten.

Why? Count People, Not Percentages

Restricting to the positives is exactly the conditioning step

P(sick | positive) = 1/11 ≈ 9%



What Just Happened

Restricting attention to the people who tested positive is exactly the move from the conditional-probability definition:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

The 11 positives are the new sample space B; the single sick person among them is $A \cap B$. The rarity of the disease survives the evidence because there were so many more healthy people to misclassify.

The trap

"99% accurate" describes $P(\text{positive} \mid \text{sick})$.

You wanted $P(\text{sick} \mid \text{positive})$.

These are different numbers.

Bayes' Theorem — Two Events

Theorem (Bayes, 2 events). Let $(\Omega, \mathcal{P}(\Omega), P)$ be a probability space and $A, B \in \mathcal{P}(\Omega)$ two events with $P(B) > 0$. Then:

$$P(A \mid B) = \frac{P(B \mid A) \cdot P(A)}{P(B)}$$

posterior — what we believe after

likelihood — how well A explains B

evidence — how ordinary B is

prior — belief before the evidence

Which of the four was "99% accurate", and which one did the intuition ignore?

Proof — Two Events

From the definition of conditional probability, twice:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{P(A \cap B)}{P(A)} \cdot P(A)}{P(B)} = \frac{P(B \mid A) \cdot P(A)}{P(B)}$$

Bayes' Theorem — Multiple Hypotheses

Theorem (Bayes, multiple events). Let $A_i \in \mathcal{P}(\Omega)$ be mutually exclusive with $\Omega = A_1 \cup \ldots \cup A_k$ and $P(A_i) > 0$, and let B be an event with $P(B) > 0$. Then

$$P(A_i \mid B) = \frac{P(B \mid A_i) \cdot P(A_i)}{P(B)} = \frac{P(B \mid A_i) \cdot P(A_i)}{\sum_{j=1}^{k} P(B \mid A_j) \cdot P(A_j)}$$

The denominator is the **total probability** of the evidence: the same observation, explained by every competing hypothesis, weighted by how much we believed each one beforehand.

Proof — Multiple Hypotheses

As before, with the theorem of total probability supplying the denominator:

$$P(B) = \sum_{j=1}^{k} P(B \mid A_j) \cdot P(A_j)$$

Mini-Exercise

Same test, but now screening a **high-risk group** where 1 in 20 people is sick — the test is unchanged.

Estimate $P(\text{sick} \mid \text{positive})$ before computing it.

Mini-Exercise — Solution

Mini-Exercise — Solution

Out of 1,000: 50 sick → 49.5 true positives; 950 healthy → 9.5 false alarms.

$$P(\text{sick} \mid \text{positive}) = \frac{49.5}{49.5 + 9.5} = \frac{49.5}{59} \approx 84\%$$

The test did not change. The prior did. This is why screening decisions depend on who is screened.

Takeaway

Bayes' theorem is the arithmetic of changing your mind: **posterior ∝ likelihood × prior**. Evidence never speaks alone — it is always weighed against how plausible the hypothesis was to begin with.

Back to Lecture 01: the Model Says "Fraud"

A fraud model is **99% accurate** in both directions. One transaction in **1,000** is fraudulent. Per 100,000 transactions:

	flagged	not flagged	total
fraud	99	1	100
legitimate	999	98,901	99,900
total	**1,098**	98,902	100,000

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legitimate	999	98,901	99,900
total	**1,098**	98,902	100,000

$$P(\text{fraud} \mid \text{flagged}) = \frac{99}{1{,}098} \approx 9\%$$

> Nine in ten blocked cards belong to honest customers — from a "99% accurate" model. The label "fraud" hid the **prior**; a score would at least have carried it.

Independence

Definition (Statistical independence). Two events A and B are **independent** iff

$$P(A \cap B) = P(A) \cdot P(B)$$

If additionally $0 < P(B) < 1$, this is equivalent to

$$P(A \mid B) = P(A) \qquad \text{and} \qquad P(A \mid B) = P(A \mid \overline{B})$$

Independence means: **learning B changes nothing about A.**

The Classic Confusion

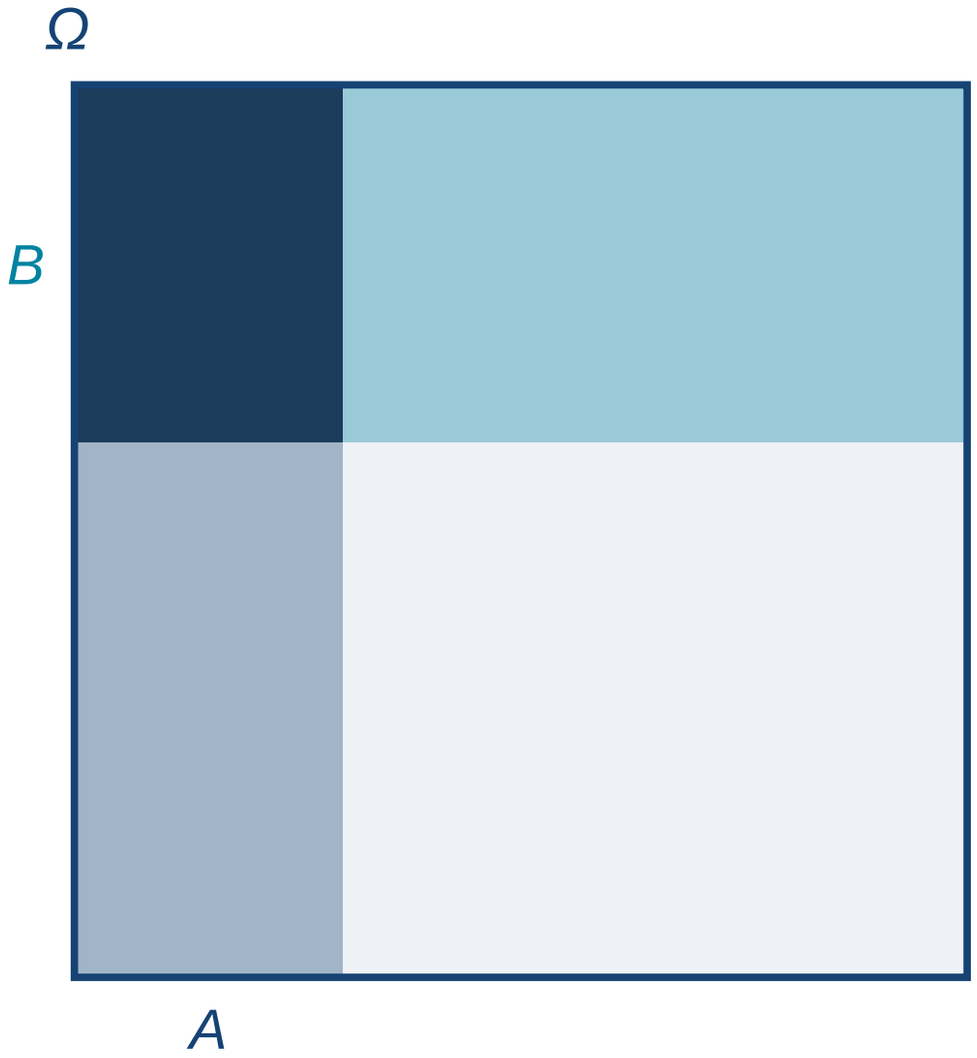
> **Statistical independence does not mean the events are disjoint.**

Disjoint events with positive probability are about as **dependent** as events can be: if A occurs, B certainly did not.

$$A \cap B = \emptyset \Rightarrow P(A \mid B) = 0 \neq P(A)$$

Independence Is About Areas, Not Overlap

$0.30 \cdot 0.40 = 0.12 = P(A \cap B) \rightarrow$ independent



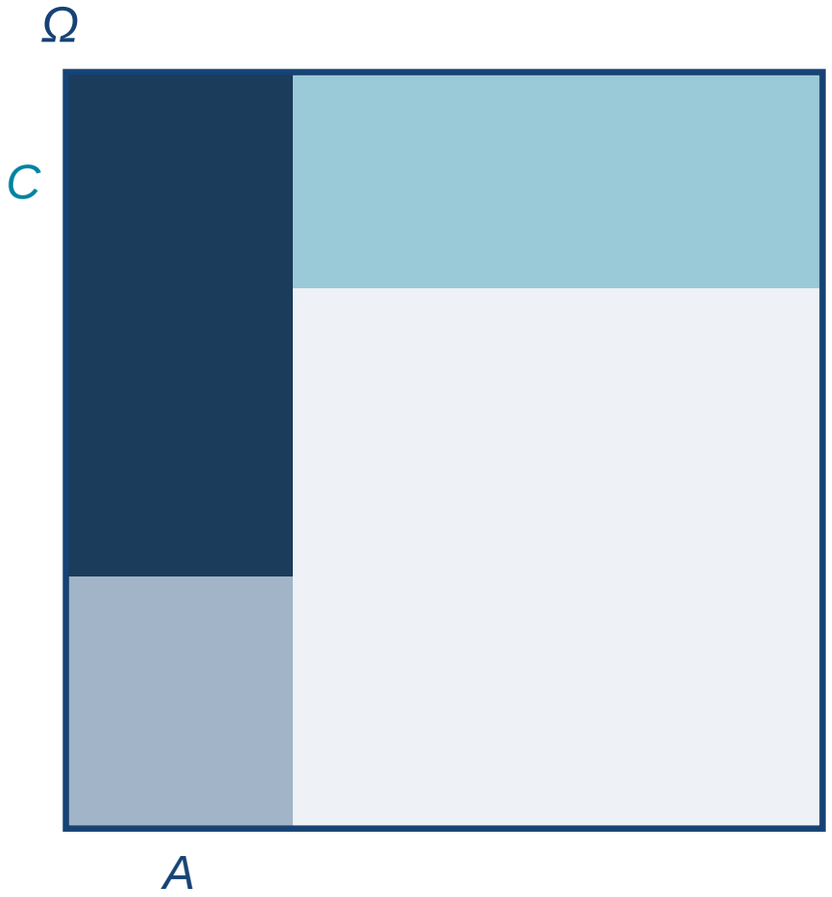
P(A) = 0.30

P(B) = 0.40

P(A ∩ B) = 0.12

The Same Picture, Dependent

0.30 · 0.40 = 0.12 ≠ 0.20 = P(A ∩ C) → dependent



P(A) = 0.30

P(C) = 0.40

P(A ∩ C) = 0.20

Here $P(A) \cdot P(C) = 0.12$ but $P(A \cap C) = 0.20$: the intersection is **larger** than independence would predict — A and C occur together more often than chance.

More Than Two Events

Events $A_1, \ldots, A_n$ are (mutually) independent iff for **every** subset of indices $i_1, \ldots, i_l$:

$$P(A_{i_1} \cap \ldots \cap A_{i_l}) = P(A_{i_1}) \cdots P(A_{i_l})$$

Pairwise independence is **not** enough — every subset must factorise.

Takeaway

> Independence is a statement about **numbers**, not pictures: $P(A \cap B)$ equals the product of the marginals. Disjoint is not independent, and identical marginals can hide either.

Module 3

From Outcomes to the Joint

> ***Guiding Question:*** *We rarely care which outcome occurred, only what it was worth — and never about one quantity alone. How do we get from a sample space to the joint distribution the three moves start from?*

Motivation: We Rarely Care About the Outcome Itself

Toss three fair coins: $\Omega = \{H, T\} \times \{H, T\} \times \{H, T\}$, eight outcomes.

Usually the interesting quantity is not *which* sequence occurred, but the **number of heads** Y:

$P(Y = 0) = \frac{1}{8}$	$P(Y = 1) = \frac{3}{8}$	$P(Y = 2) = \frac{3}{8}$	$P(Y = 3) = \frac{1}{8}$
(T, T, T)	$(H, T, T), (T, H, T), (T, T, H)$	$(H, H, T), (T, H, H), (H, T, H)$	(H, H, H)

Y compresses eight outcomes into four numbers — and the probabilities come along automatically.

Definition

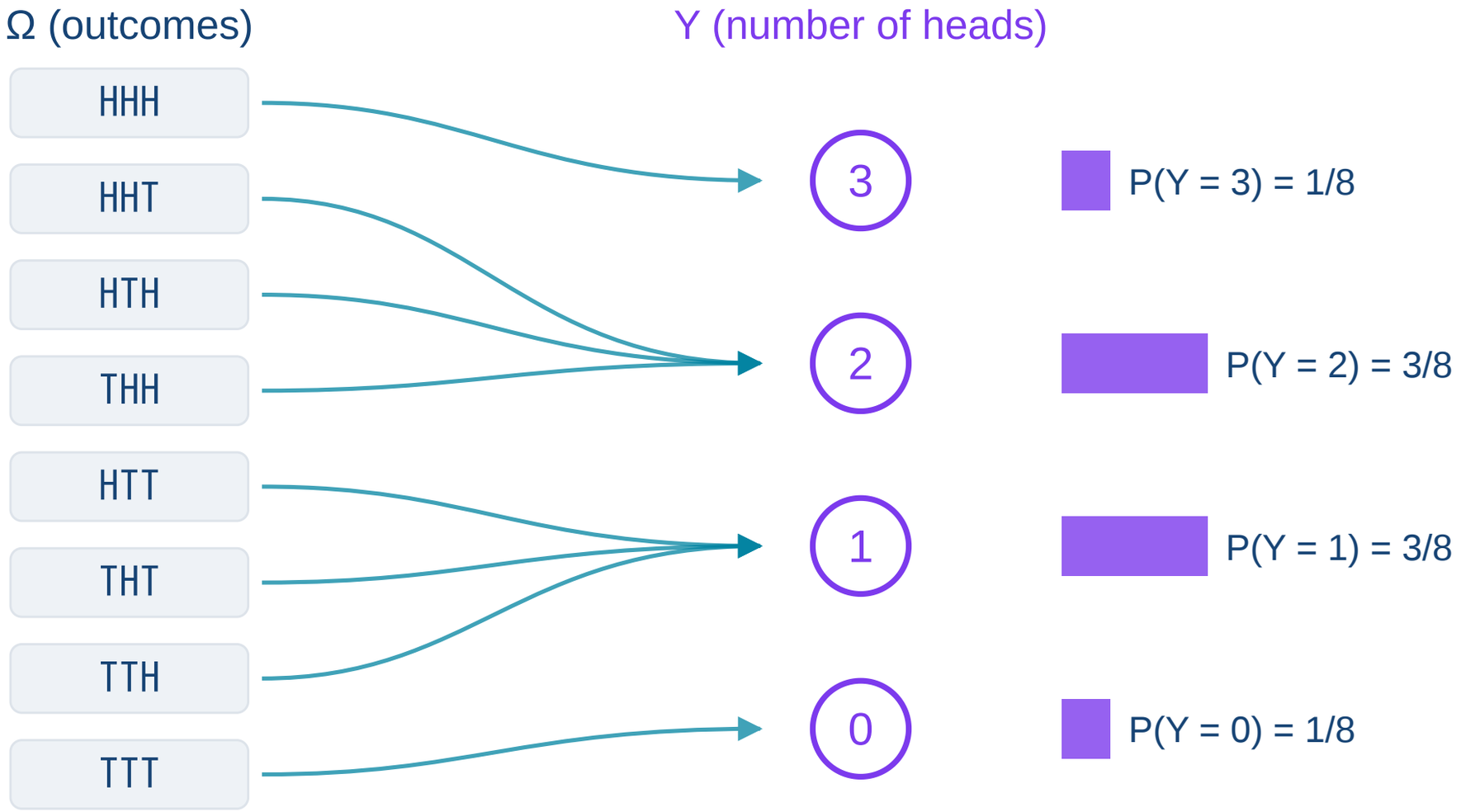
Definition (Random variable). A random variable X maps outcomes to a measurable range E: $X : \Omega \to E$. Its probabilities are inherited from the underlying probability space:

$$P(X \in S) = P(\{\omega \in \Omega \mid X(\omega) \in S\})$$

A random variable is neither random nor a variable — it is a **function** on the sample space. The randomness lives in ω, not in X.

The Function, Drawn

Three arrows arrive at 1 and at 2 — which is the whole shape



Examples

Two coins

$\Omega = \{HH, HT, TH, TT\}$

X = number of heads

People

$\Omega = \{\text{person}_1, \text{person}_2, \ldots\}$

X = height of the person

Two dice

$\Omega = \{\langle n_1, n_2 \rangle\}$

$X(\omega) = n_1 + n_2$

Discrete and Continuous

Discrete random variables

Countable range, usually $E \subseteq \mathbb{N}$.

- number of children of a person
- sum of two dice

Continuous random variables

Real-valued range, $E = \mathbb{R}$.

- height of a person
- horsepower of a car

The distinction decides *how* we may describe the probabilities — with a mass function or with a density.



A Random Variable With Infinitely Many Values

Flip a biased coin ($P(H) = p$) until the first head. Let X be the number of flips needed:

$$P(X = 1) = p, \quad P(X = 2) = (1-p)p, \quad P(X = 3) = (1-p)^2 p$$

$$P(X = n) = (1-p)^{n-1} p$$

Countably infinite range, and the probabilities still sum to 1.

Takeaway

A random variable is a **function from outcomes to numbers** that carries the probability measure with it. It lets us stop talking about sample spaces and start talking about quantities — which is what the rest of the course does.

Probability Mass Function

Definition (PMF). For a discrete random variable X, the **probability mass function** is

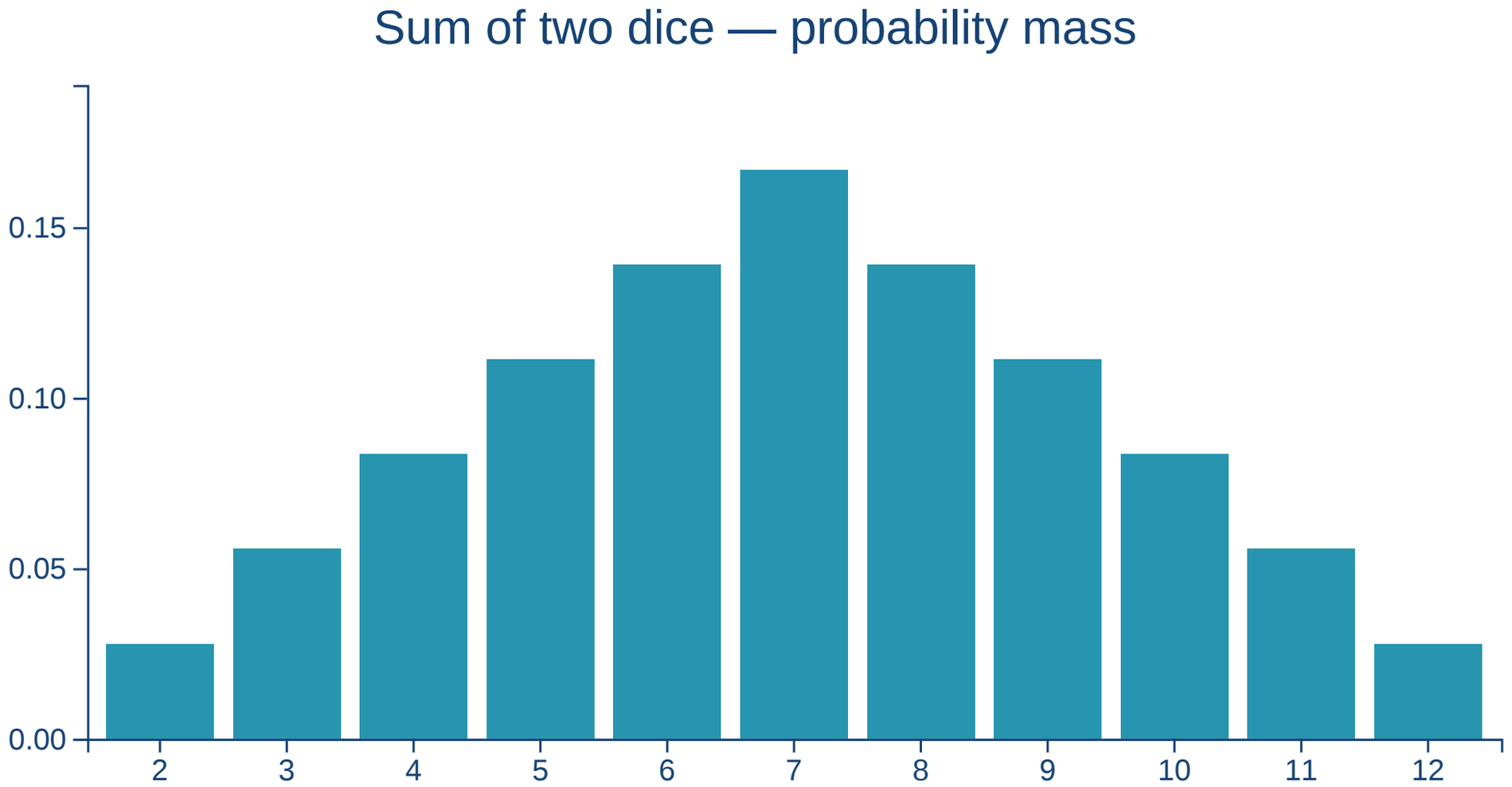
$$p(a) = P(X = a)$$

$p(a) > 0$ for at most countably many a

$\sum_i p(a_i) = 1$

p is itself a probability measure

Example: Sum of Two Dice



The triangular shape is counting, drawn: six of the 36 outcomes give 7, only one gives 2.

Expectation

Definition (Expected value).

$$E[X] = \sum_i a_i \cdot p(a_i)$$

- Frequentist reading: the **average outcome** if the experiment were repeated infinitely often
- The first moment: $\mu = E[X^1]$; the first *central* moment $E[X - E[X]]$ is always 0

For a real function g:

$$E[g(X)] = \sum_i g(a_i)p(a_i)$$

and for constants a, b:

$$E[aX + b] = aE[X] + b$$

Variance

Definition (Variance).

$$\mathrm{Var}(X) = E\big[(X-\mu)^2\big] = E[X^2] - \big(E[X]\big)^2$$

- a measure of **dispersion** around the mean — larger variance, larger typical deviation
- the second central moment
- $\mathrm{Var}(aX+b) = a^2\mathrm{Var}(X)$
- $\sigma = \sqrt{\mathrm{Var}(X)}$ is the **standard deviation**, in the same unit as X

Takeaway

A discrete distribution is a PMF; expectation and variance compress it into **where it sits** and **how much it spreads**.

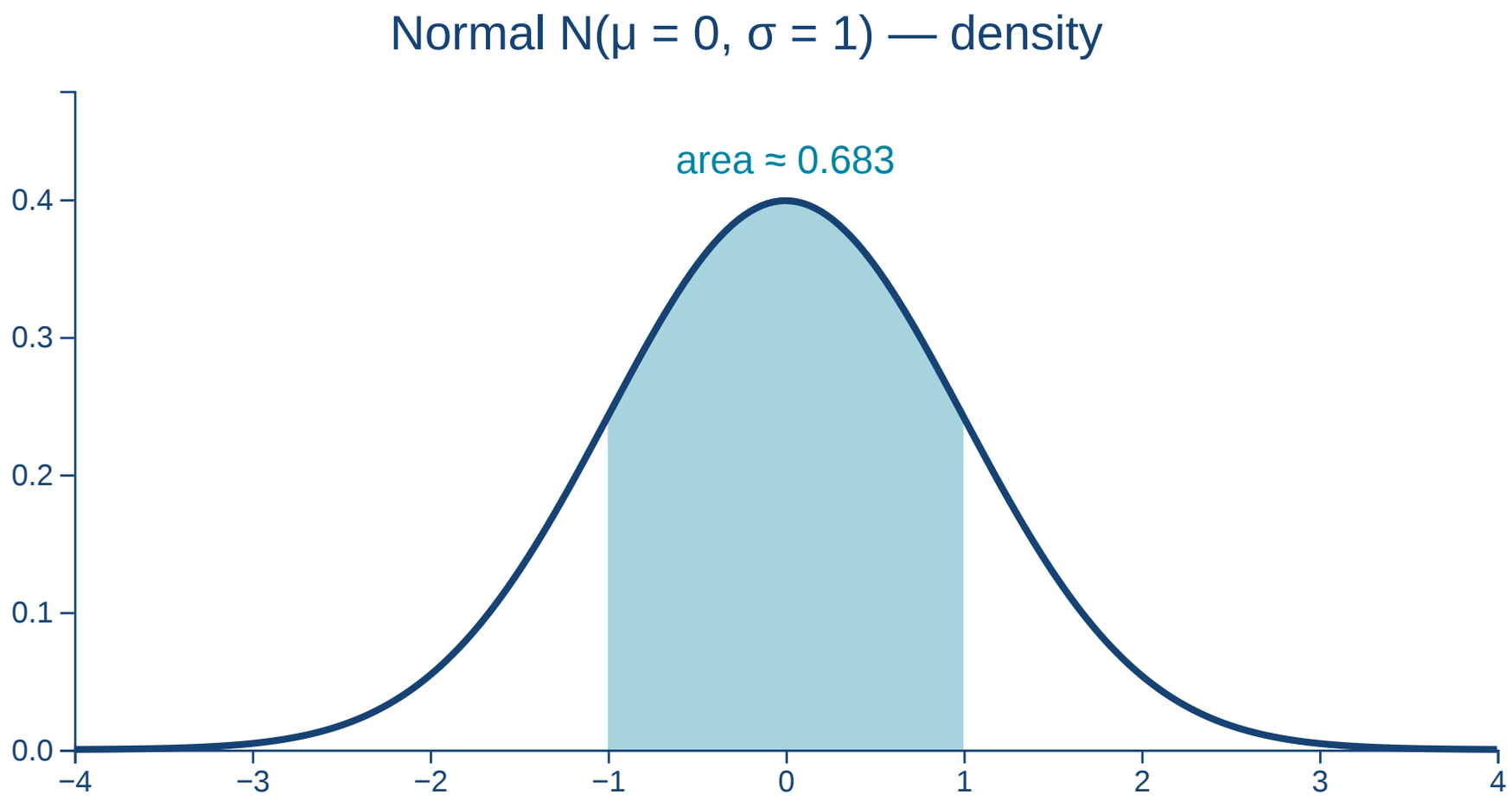
From Mass to Density

Definition (PDF). A continuous random variable X has a **probability density function** f with

$$P(b \leq X \leq c) = \int_b^c f(a)da, \qquad \int_{-\infty}^{\infty} f(a)da = 1$$

$f(a)$ is **not** a probability — it is probability *per unit interval*. Consequently $f(a) > 1$ is perfectly possible, and $P(X = a) = 0$ for every single value a.

Probability Is Area Under the Curve



The shaded region is $P(-1 \leq X \leq 1)$ — the same "probability as area" intuition from the axioms unit, now with a curve instead of a Venn diagram.

Expectation and Variance — Sums Become Integrals

Discrete

$$E[X] = \sum_i a_i p(a_i)$$

$$E[g(X)] = \sum_i g(a_i) p(a_i)$$

Continuous

$$E[X] = \int_{-\infty}^{\infty} a f(a) da$$

$$E[g(X)] = \int_{-\infty}^{\infty} g(a) f(a) da$$

The rules are unchanged: $E[aX + b] = aE[X] + b$, and $\mathrm{Var}(X) = E[(X - \mu)^2]$.

Two Basic Continuous Distributions

Uniform on $[a, b]$

$$f(x) = \frac{1}{b-a} \text{ for } x \in [a, b]$$

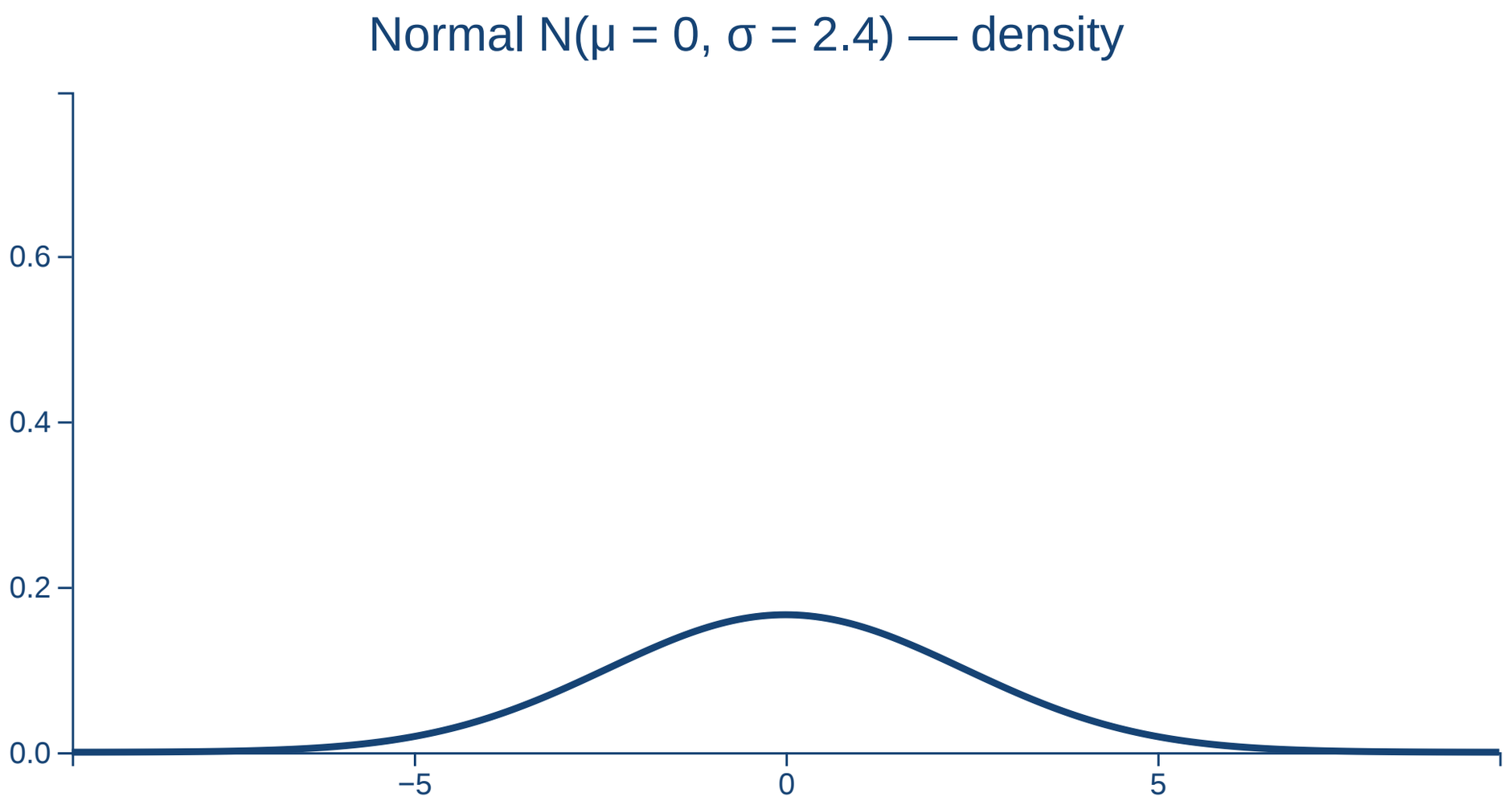
Every interval of equal length is equally likely — the Laplace assumption, continuous version.

Normal $N(\mu, \sigma^2)$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$E[X] = \mu$, $\mathrm{Var}(X) = \sigma^2$; $N(0, 1)$ is the **standard** normal.

What σ Does to the Shape



The peak falls as the curve widens — the area is always 1.

Why the Normal Is Everywhere

> The **central limit theorem**: under mild conditions, the average of n independent observations is approximately normal for large n — whatever the original distribution was.

This is why the normal is the default when the true distribution is unknown — and why a sum of many independent coin flips, the binomial, looks bell-shaped.

Other Distributions You Will Meet

- **Exponential** — time until an event occurs (e.g. until the next earthquake)
- **Gamma** — time until n events have occurred
- **Beta** — values in $[0, 1]$, e.g. an unknown probability
- **Dirichlet** — distributions *over* probabilities

Continuous distributions replace mass with **density** and sums with **integrals**. Probability remains area, expectation remains a weighted average, and the normal is the default shape because of the central limit theorem.

More Than One Random Variable

Univariate	Bivariate	Multivariate
one random variable	two random variables	three or more

A **joint distribution** describes the outcomes of several random variables *at the same time* — which is what every dataset with more than one column is.

Joint Mass and Density

Discrete — joint PMF

$$p_{X,Y}(a,b) = P(X = a, Y = b)$$

Continuous — joint PDF

$$P((X,Y) \in R) = \iint_R f_{X,Y}(a,b)dadb$$

Notational convention from here on. We stop distinguishing discrete from continuous variables and simply write $p(a,b,c,\ldots)$ for the probability that $\{X_1, X_2, X_3, \ldots\}$ is in state $a, b, c, \ldots$

Marginalisation

Marginalisation eliminates a random variable by summing over all its states. For $p(x_1, \ldots, x_n)$, a **marginal** over a subset is

$$p(x_1) = \sum_{x_2} \cdots \sum_{x_n} p(x_1, x_2, \ldots, x_n)$$

"Forget about a variable, but keep its influence correctly accounted for."

Example: Boys and Girls

15% of families have no children, 20% have one, 35% have two, 30% have three. Each child is independently a boy or a girl with probability $\frac{1}{2}$.

Let B = number of boys, G = number of girls.

$p(B, G)$	$G = 0$	$G = 1$	$G = 2$	$G = 3$	$p(B)$
$B = 0$	.15	.10	.0875	.0375	.375
$B = 1$	.10	.175	.1125	0	.3875
$B = 2$	.0875	.1125	0	0	.2
$B = 3$	.0375	0	0	0	.0375
$p(G)$	.375	.3875	.2	.0375	1

The margins of the table *are* the marginals — which is where the name comes from.

Covariance

$$\operatorname{cov}(A, B) = E\big[(A - E[A])(B - E[B])\big]$$

$\operatorname{cov} > 0$
A large when B large — positively related

$\operatorname{cov} < 0$
A large when B small — negatively related

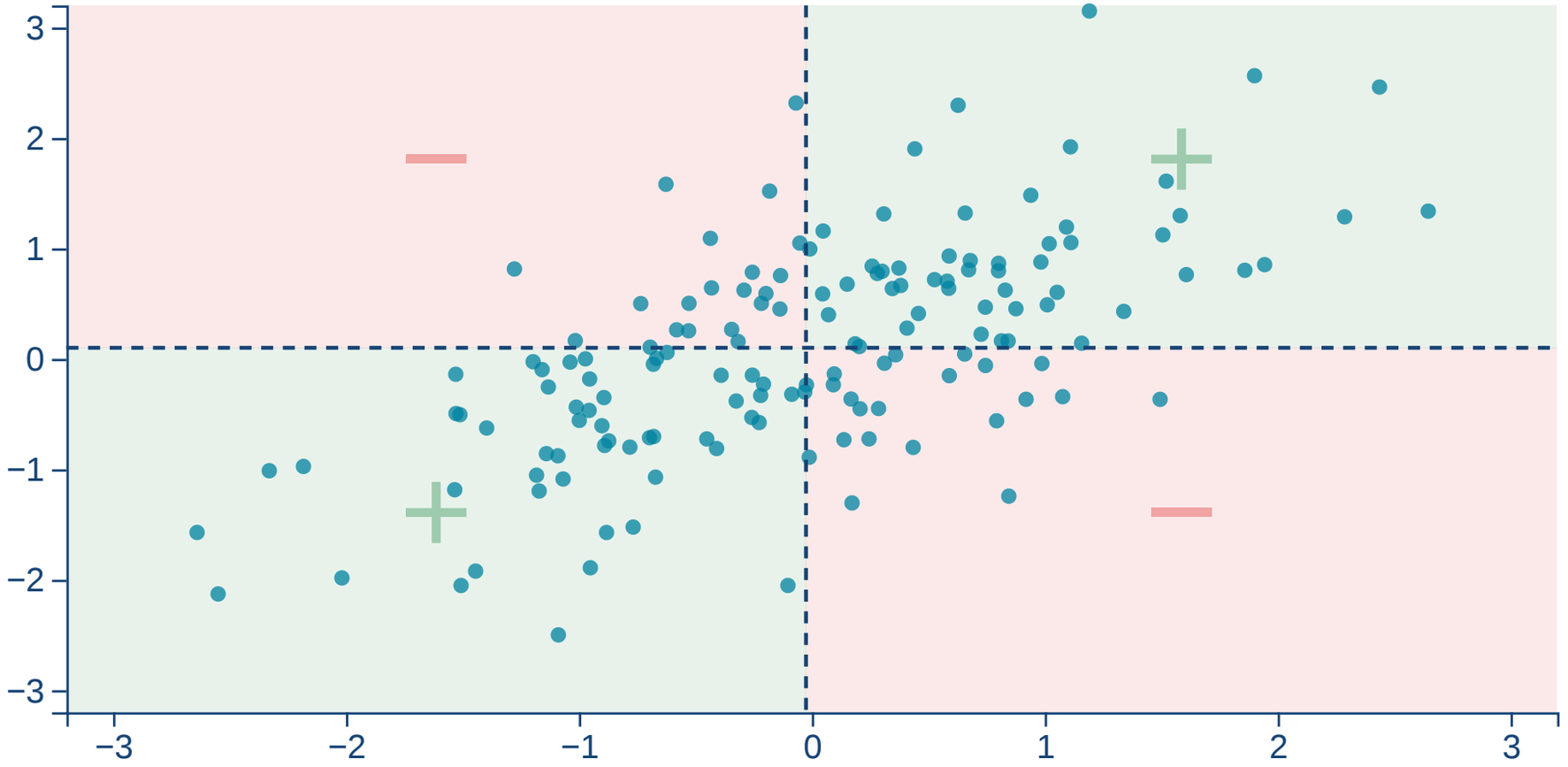
$\operatorname{cov}(A, A) = \operatorname{Var}(A)$
variance is self-covariance

Independence implies zero covariance. **The converse is false** — covariance only detects *monotone linear* association.

Covariance Sees Only One Kind of Structure

Slide ρ, then switch the relation. Predict the covariance first.

cov(X, Y) = 0.62



Predict first: with $Y = X^2 - 1$, every point is fixed by its x. The covariance?

Takeaway

The joint distribution holds everything; **marginalisation** sums a variable away, **conditioning** restricts, and **covariance** summarises linear co-variation. Zero covariance is weaker than independence.

Module 4

Data, Statistics and a Parameter

> ***Guiding Question:*** *A model has a parameter nobody told us, and all we have is a table. What does the table say about the parameter — and how sure may we be?*

Four Attribute Types

Nominal

categorical, qualitative

Labels only: eye colour, ZIP code, ID.

Meaningful: $=$, $\neq$

Ordinal

categorical, qualitative

Ordered labels: grades, rankings, "small/medium/large".

Meaningful: $<$, $>$

Interval

numeric, quantitative

Differences are meaningful, the zero point is arbitrary: temperature in °C, calendar dates.

Meaningful: $+$, $-$

Ratio

numeric, quantitative

A true zero, so ratios are meaningful: length, mass, counts, duration.

Meaningful: $\times$, $\div$

Why the Type Decides the Method

The classic error

Averaging a nominal attribute. The mean of ZIP codes 10115 and 81667 is a number, and it is meaningless.

The subtle one

20 °C is not "twice as warm" as 10 °C — interval scales have no true zero, so ratios do not transfer.

The attribute type determines which **statistics**, which **distance measures** and which **plots** are permissible. Every method in this course depends on it.

Mini-Exercise

A customer table records satisfaction (one to five stars) and year of birth. Which scale is each — and whose **mean** would you not report?

Mini-Exercise

A customer table records satisfaction (one to five stars) and year of birth. Which scale is each — and whose **mean** would you not report?

Satisfaction is **ordinal**: its mean assumes equal gaps between the stars — report the median. Year of birth is **interval**: the mean year is fine, but ratios are not ("born twice as late" means nothing); differences such as age are fine.

Takeaway

> An attribute's **scale of measurement** licenses the operations you may perform on it. Choosing a statistic before checking the scale is the most common silent error in data analysis.

A Table Is a Sample From a Joint Distribution

customer	payment	satisfaction	born	income (k€)	returned?
1	card	4	1987	52	yes
2	invoice	2	1995	31	no
3	card	5	1979	78	no
…	…	…	…	…	…

Each column is a random variable

$X_{\text{payment}}, X_{\text{satisf.}}, \ldots$ — a function from customers to values, exactly as a random variable was defined.

Each row is one draw

The whole table is n draws from one joint distribution $p(x_1, \ldots, x_d)$ that nobody has written down.

The Scale Suggests the Model

attribute	scale	a first model
returned?	nominal, two values	**Bernoulli**(p) — one parameter
payment	nominal, k values	**categorical**$(p_1, \ldots, p_k)$, $\sum_j p_j = 1$
satisfaction	ordinal	categorical — the order is not used, or an ordinal model
born	interval	normal, on a restricted range
income	ratio, positive, skewed	log-normal, or normal on $\log$ income

> The scale limits the model as it limits the statistics: a normal distribution for ZIP codes is as meaningless as their mean.

What "i.i.d." Claims

The standard reading of a table rests on one assumption: the rows are **independent and identically distributed**.

Independent

One customer's row tells us nothing about another's.

Identically distributed

Every row comes from the **same** joint distribution.

Which of these datasets break it: daily sales of one shop · two visits by the same customer · pixels of one image · a survey sent to a random sample?

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Only the random survey is plausibly i.i.d. Time series, repeated measurements and spatial data all violate it — the arrangement *is* information.

Takeaway

A dataset is n **draws from an unknown joint distribution**: columns are random variables, rows are draws, and the scale of each column suggests its model. Reading a table this way rests on an assumption — **i.i.d.** — that should be checked, not presumed.

Bernoulli and Binomial

Bernoulli(p) — one trial

$X = 1$ success, $X = 0$ failure

$$E[X] = p, \quad \mathrm{Var}(X) = p(1-p)$$

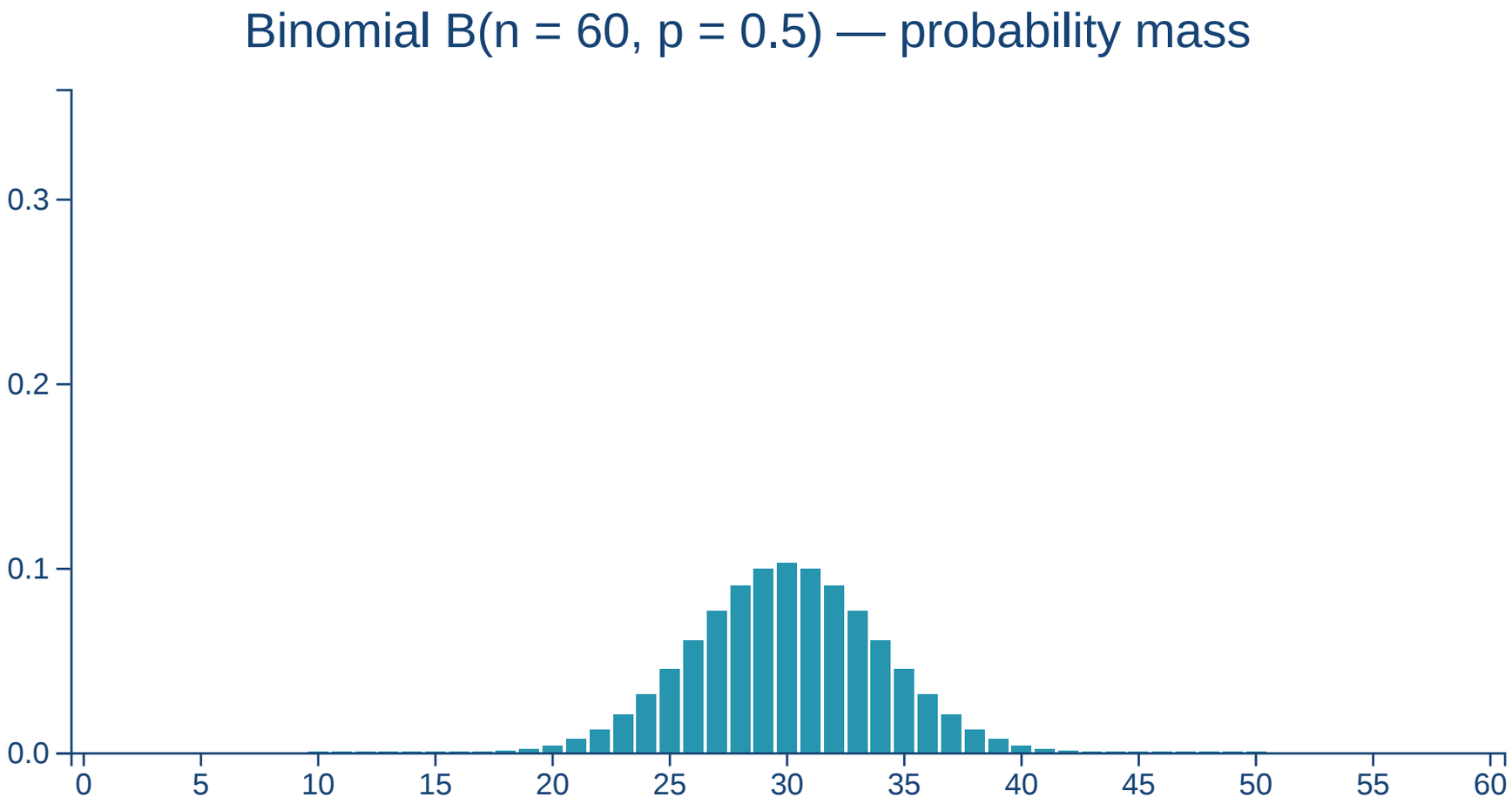
Binomial $B(n, p)$ — n independent trials

$$p(k) = \binom{n}{k} p^k (1-p)^{n-k}$$

$$E[X] = np, \quad \mathrm{Var}(X) = np(1-p)$$

$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}$ counts the sequences with k successes; the PMF multiplies it by the probability of one such sequence — combinatorics and independence in a single line.

How the Shape Depends on the Parameters



Predict before each click: as n grows, does the distribution get wider or narrower — and what happens to its shape?

Mini-Exercise

A student guesses all **10** questions of a quiz, each with **4** options.

How many correct answers should they expect — and is 5 or more likely?

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$B(10, 0.25)$: on average $10 \cdot 0.25 = 2.5$ correct. Five or more is twice the expectation — unlikely (about 8%).

Takeaway

A yes/no attribute is a **Bernoulli** variable with one parameter p; counting the yeses in n independent trials gives a **binomial**. Both are counting in probabilistic clothing — and p is the number data will have to tell us.

We Never Know the Distribution

Up to now we assumed a distribution was given. In practice it is **unknown**, and we must estimate its properties from data.

Population
The entire group — all humans on earth, all transactions, all documents.
Its properties (μ, σ^2) are the truth we cannot observe.

Sample
The observations we actually have.
Its properties ($\overline{x}$, s^2) are what we compute; estimates of the population's ($\overline{x}$, $\hat{\sigma}^2$) are built from them.

Descriptive statistics summarise a sample, usually while assuming something about the underlying distribution.

What We Summarise

Central tendency

1st order moment
sample mean, median

Dispersion

2nd central moment
variance, standard deviation, interquartile range

Shape

3rd central moment
skewness — or, without moments, quantiles

Univariate statistics describe one random variable; **multivariate** statistics describe $X_1, \ldots, X_d$ together.

Means

Sample mean

$$\overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

Weighted mean

$$\overline{x}_w = \frac{\sum_i w_i x_i}{\sum_i w_i}$$

Geometric mean — for rates and ratios

$$\left(\prod_i x_i\right)^{1/n}$$

Harmonic mean — for averaging rates

$$\frac{n}{\sum_i 1/x_i}$$

Trimmed mean: drop the k smallest and largest values, then average — a cheap way to buy robustness.

Median

The **median** is the middle value of the sorted sample: for odd n the $\frac{n+1}{2}$-th value, for even n the average of the two middle values.

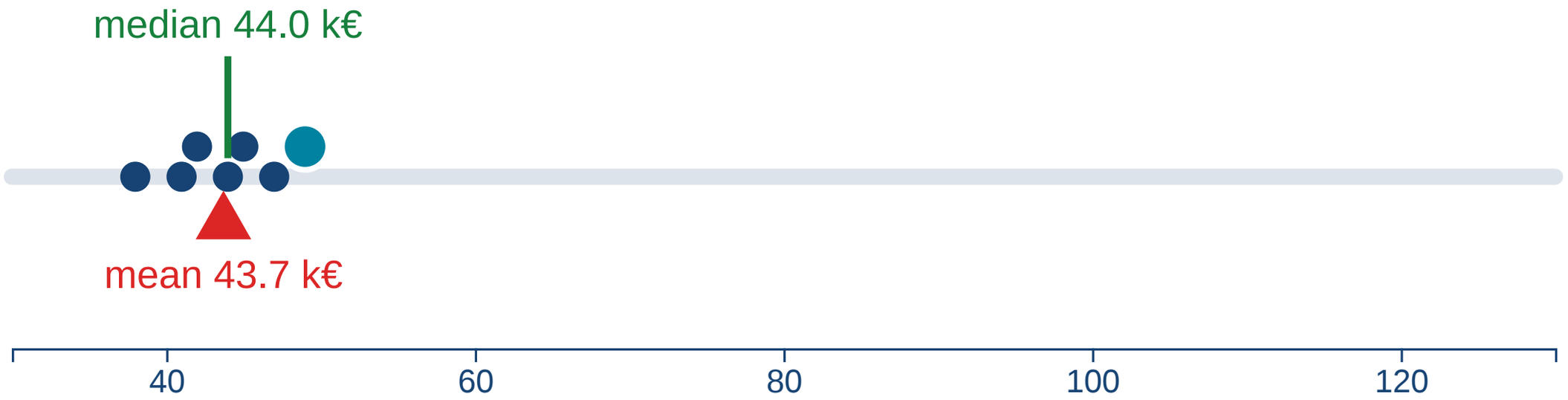
Robust against outliers — one extreme value moves the median by at most one position

Costlier to compute: requires sorting, or an interval approximation over binned data

For binned data, the median is interpolated within its interval from the cumulative counts, the interval's lower border and width.

One Value, Two Very Different Reactions

Drag the light point — which marker follows it?



Commit first: drag the top salary to the far right. Which marker follows?

Mini-Exercise

Salaries in a small firm (in k€): 38, 41, 42, 44, 45, 47, 260.

Compute the mean and the median. Which number should appear in a recruitment ad, and which one describes a typical employee?

Mini-Exercise — Solution

Mini-Exercise — Solution

Mean ≈ 73.9, median $= 44$.

The single large value pulls the mean above **every** salary except one. The median describes the typical employee; the mean describes the payroll divided by headcount. Both are correct — they answer different questions.

Dispersion

Dispersion describes how stretched or squeezed the data is around its centre.

Sample variance — the variance *of the observed data*

$$s^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \overline{x})^2$$

Standard deviation $s = \sqrt{s^2}$, in the unit of the data.

Computable in **one pass** over the data, $O(n)$, using

$$\sum x_i^2 \text{ and } \sum x_i$$

— which matters when the data does not fit in memory.

Why the Denominator Changes

The **population variance** σ^2 is the variance over the whole population. If the true mean μ is unknown, we must estimate it — and using $\overline{x}$ in its place makes the deviations systematically too small.

$$\hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \overline{x})^2$$

Dividing by $n-1$ compensates: one degree of freedom was spent estimating the mean from the same data.

Naming is not uniform: many textbooks, and pandas, call the $n-1$ version s^2; NumPy's `var` divides by n unless told otherwise.

Takeaway

s^2 describes **the sample**; $\hat{\sigma}^2$ estimates **the population**. The $n-1$ is not a convention — it is the price of having estimated the mean from the same data.

You Computed a Number. Which Number Did You Want?

Seven salaries gave a sample mean of $\overline{x} = 73.9$ k€.

What is the probability that the population mean μ is exactly 73.9 k€?

You Computed a Number. Which Number Did You Want?

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What is the probability that the population mean μ is exactly 73.9 k€?

Zero. $\overline{x}$ is not μ — it is one **estimate** of it, computed from the seven people who happened to be in the sample.

The honest question is not *what is* μ but **how far from** μ **can this number be?**

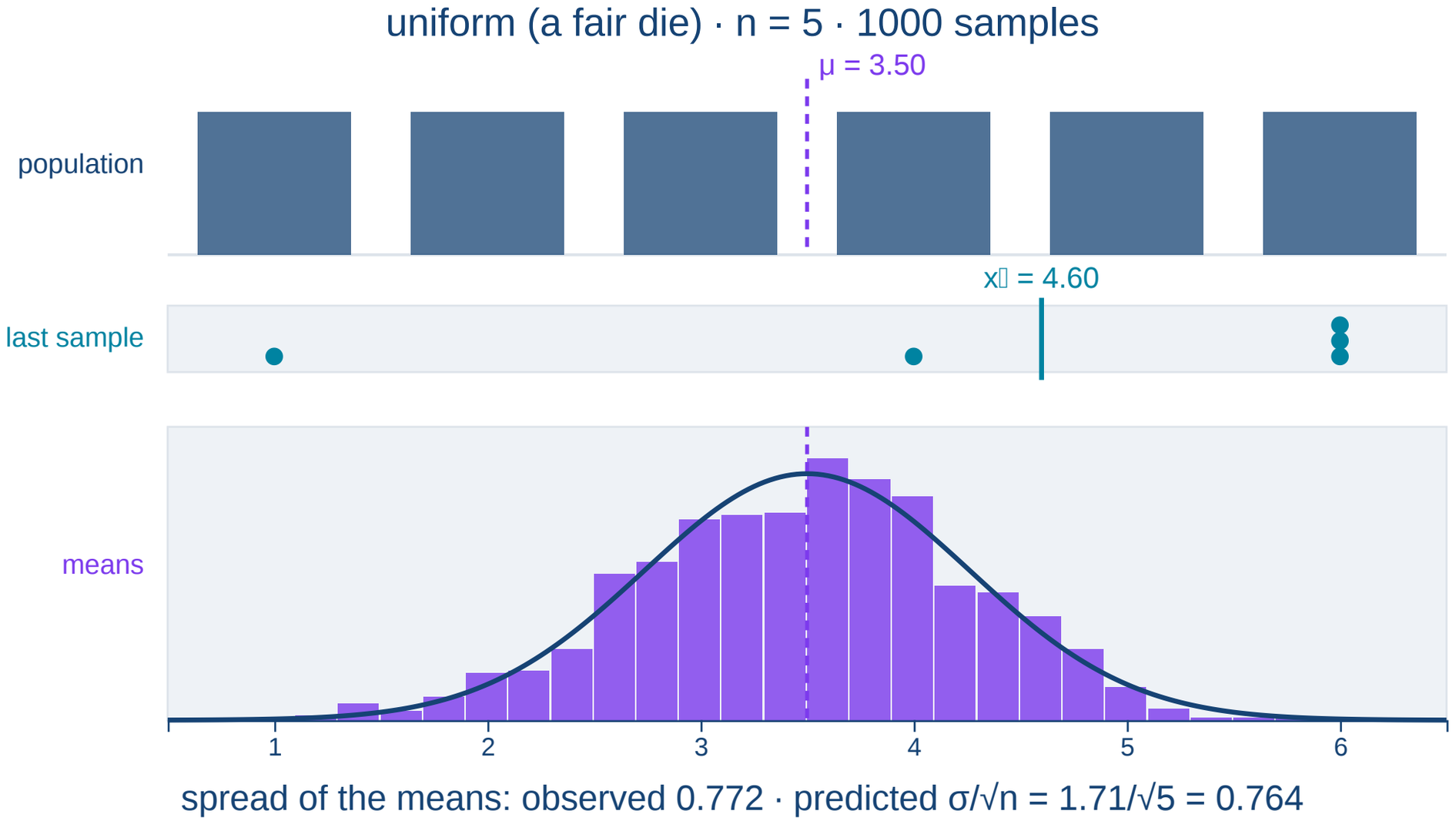
Estimator and Estimate

Definition. An **estimator** is a function of the sample — $\hat{\theta} = g(X_1, \ldots, X_n)$ — used to approximate an unknown quantity θ of the population. The number it takes on for one particular sample is an **estimate**.

The sample is random, so the estimator is a **function of random variables**: $\hat{\theta}$ is itself a random variable, with a distribution of its own.

$\overline{x}$, s^2 and $\hat{\sigma}^2$ from descriptive statistics are all estimators. We used them without saying what kind of object they are.

An Estimator Has a Distribution



Three Things the Picture Shows

- → The means pile up **around** μ — the estimator is not systematically off
- → The pile is **narrower than the population** — averaging cancels noise
- → The pile is **bell-shaped**, even though the population was not

> **Definition (Sampling distribution).** The distribution of an estimator $\hat{\theta}$ over all possible samples of size n. $\hat{\theta}$ is **unbiased** if $E[\hat{\theta}] = \theta$.

Use the Widget: Change the Population

Set the population to **skewed** and then to **bimodal**, keeping $n = 5$.

Which of the three observations survive a population that is not symmetric?

Use the Widget: Change the Population

Set the population to **skewed** and then to **bimodal**, keeping $n = 5$.

Which of the three observations survive a population that is not symmetric?

All three. The shape of the sampling distribution stops depending on the shape of the population as soon as n is more than a handful — which is why one formula can cover cases we never inspect.

Standard Error

Definition (Standard error). The standard deviation of an estimator's sampling distribution. For the sample mean of independent draws:

$$\mathrm{SE}(\overline{X}) = \frac{\sigma}{\sqrt{n}}$$

It is measured in the unit of the data and answers "how far off is a typical estimate", where σ answers "how far apart are typical *observations*".

σ and SE are different quantities. More data does nothing to σ — the world stays as variable as it was — and shrinks SE.

Precision Costs Data

Multiply the data by…	…and the standard error is multiplied by
2	0.71
4	**½**
25	⅕
100	**1⁄10**

The $\sqrt{n}$ is why the last digit of an estimate is so expensive: halving the error costs four times the data, and a tenfold improvement costs a hundredfold.

Two Theorems We Have Already Seen Happen

Law of large numbers

$\overline{X}_n \to \mu$ as $n \to \infty$.

The estimate converges on the truth — the pile keeps narrowing.

Central limit theorem

$\overline{X}_n$ is approximately $N(\mu, \sigma^2/n)$ for large n.

And we know the *shape* of what is left.

The first says the error goes away; the second says how much of it is left after n observations. Only the second lets us put a number on our ignorance.

From a Number to an Interval

If $\overline{X} \approx N(\mu, \sigma^2/n)$, then for 95% of samples $\overline{X}$ lands within $1.96\,\mathrm{SE}$ of μ — and therefore μ lands within $1.96\,\mathrm{SE}$ of $\overline{X}$.

$$\overline{x} \pm 1.96 \cdot \frac{\sigma}{\sqrt{n}}$$

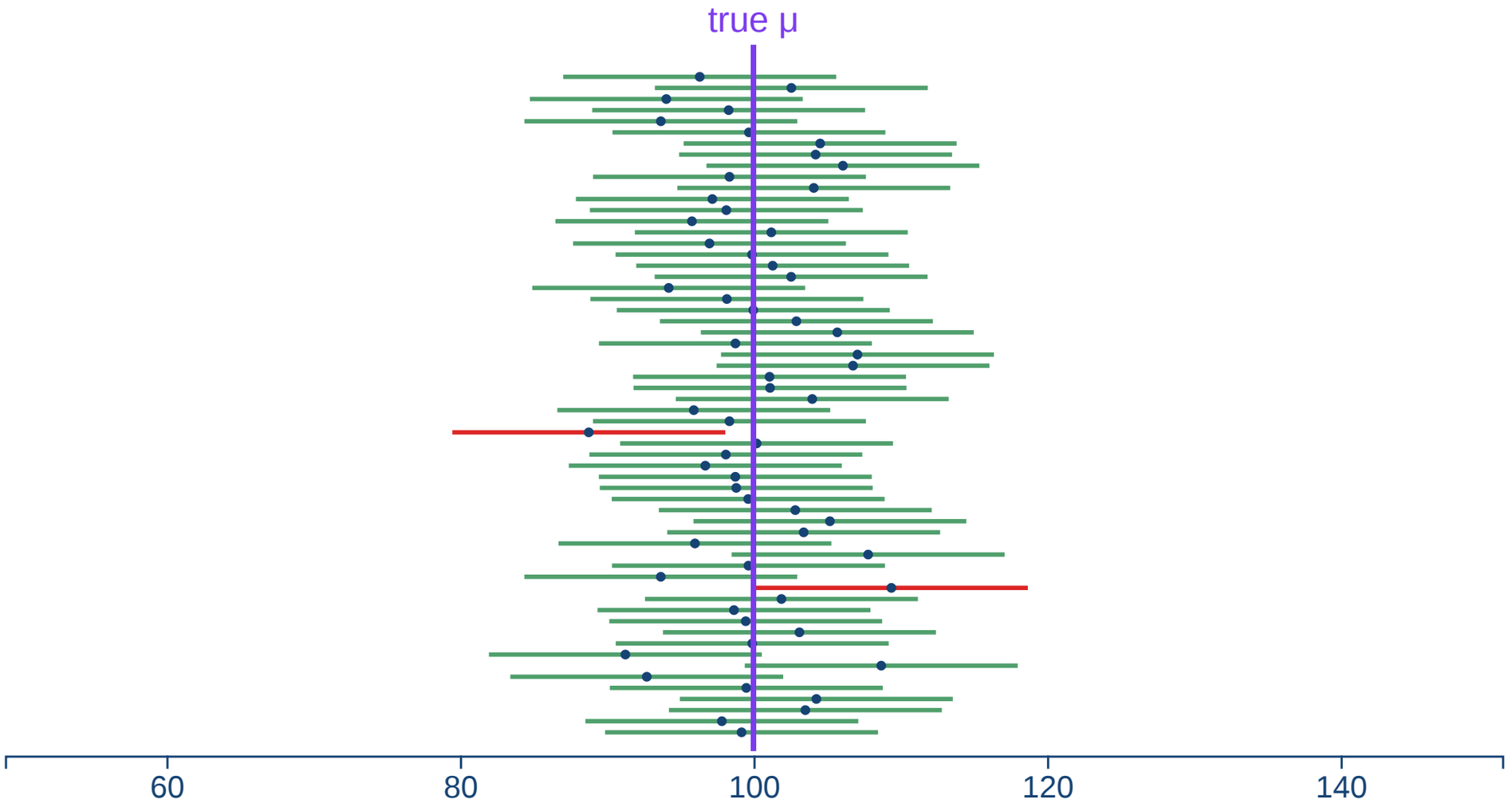
the estimate — our best single guess

how confident we insist on being

how noisy the estimate is

What Does "95% Confident" Mean?

Change the level or n — which number follows, and which does not?



58 of 60 intervals cover μ (97%) — nominal 95%

The Sentence Almost Everyone Gets Wrong

WHAT IT DOES NOT MEAN

"There is a 95% probability that μ lies in **this** interval."

μ is a fixed number. It is either in this interval or it is not — there is no probability left to have.

WHAT IT MEANS

"95% of the intervals produced **by this procedure** contain μ."

The probability is a property of the recipe, evaluated before the data is seen.

Where This Leaves Us

Two kinds of uncertainty meet in one formula. We can now measure one of them.

- → σ — how variable the world is. **Aleatoric.** Unchanged by data.
- → $\mathrm{SE} = \sigma/\sqrt{n}$ — how badly we know our own estimate. **Epistemic.** Shrinks with data.

"More data helps" was a slogan.
It is now a rate: $1/\sqrt{n}$.

Mini-Exercise

A shop timed $n = 100$ customers at a counter: $\overline{x} = 180$ s, and past records give $\sigma = 60$ s. What is the standard error — and how many customers would halve it?

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$\mathrm{SE} = 60/\sqrt{100} = 6$ s, so the 95% interval is about 180 ± 12 s. Halving the SE needs twice the $\sqrt{n}$ — **four times the data**, $n = 400$. $\sigma = 60$ s does not change: customers stay as variable as they were; only our knowledge of their average improves.

Takeaway

A statistic is a **random variable**. Its distribution over repeated samples is the sampling distribution; its spread is the **standard error**, $\sigma/\sqrt{n}$. A confidence interval reports that spread — and its percentage describes the **procedure**, not the one interval in front of you.

An Unknown Number, Ten Observations

A new feature went live. Of the first **10 users**, **3 came back**.

What is p, the probability that a user comes back — and how sure are we?

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WHAT THE LAST UNIT CAN SAY

$\hat{p} = 0.3$, standard error ≈ 0.14, a 95% interval of roughly $[0.02, 0.58]$ — about the *procedure*, not about p.

WHAT WE WOULD LIKE TO SAY

"p is probably between 0.1 and 0.6, most likely around 0.3" — a statement **about** p.

Conditioning on the Data — for a Parameter

Treat the unknown p as a random variable θ and apply **Bayes' theorem**, with θ in place of the hypothesis and the data D as the evidence:

$$p(\theta \mid D) = \frac{p(D \mid \theta) \cdot p(\theta)}{p(D)}$$

posterior — belief about θ after the data

likelihood — how probable the data is for each θ

evidence — normalises, does not depend on θ

prior — belief about θ before

For i.i.d. yes/no data with h yeses in n trials: $p(D \mid \theta) = \theta^h(1-\theta)^{n-h}$ — read as a function of θ, with the data fixed.

A Prior That Behaves Like Counts

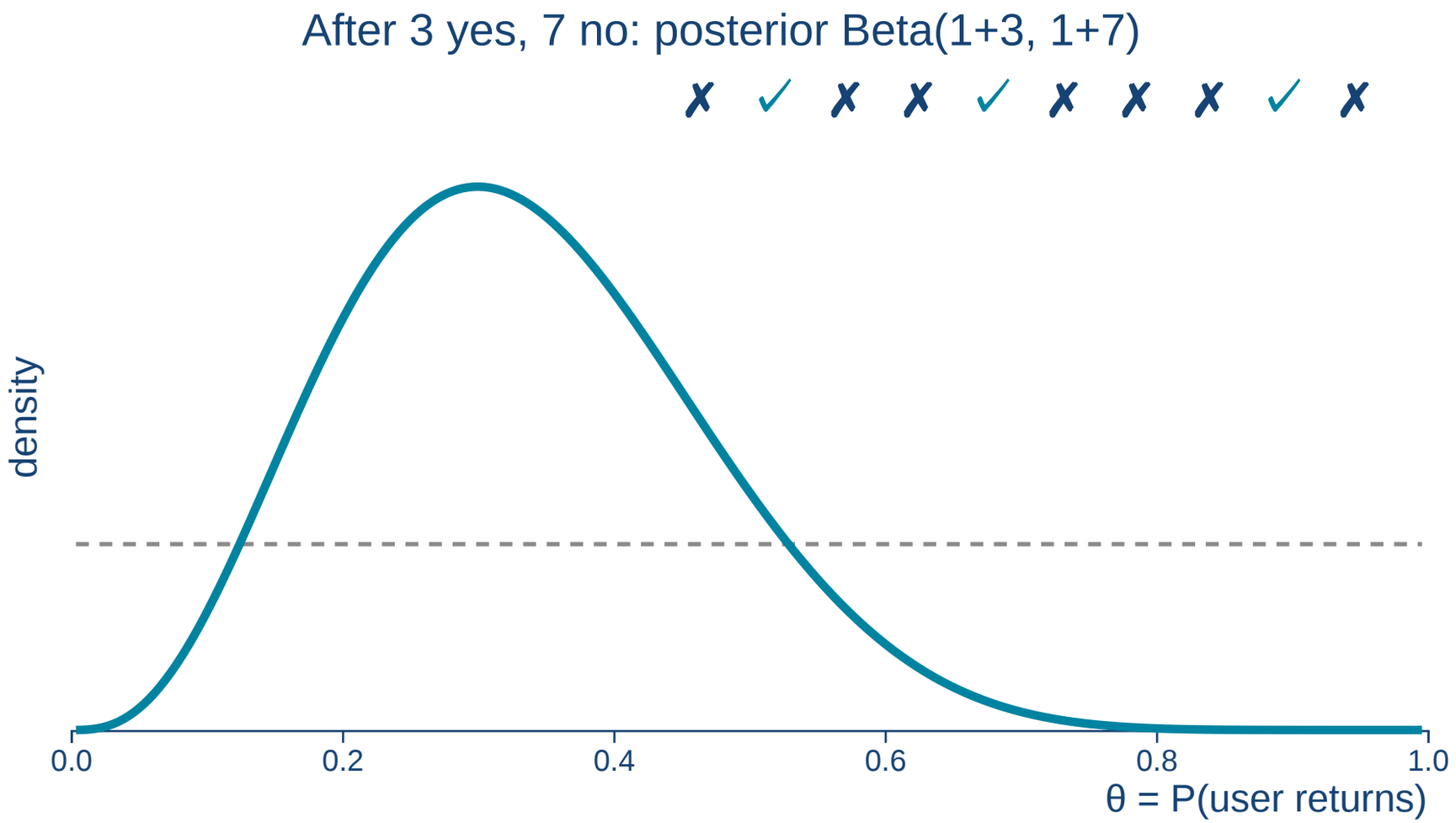
Beta prior. $p(\theta) \propto \theta^{a-1}(1-\theta)^{b-1}$ on $[0,1]$, written $\mathrm{Beta}(a,b)$. $\mathrm{Beta}(1,1)$ is flat: every p equally plausible.

Multiply by the likelihood $\theta^h(1-\theta)^{n-h}$:

$$p(\theta \mid D) \propto \theta^{a+h-1}(1-\theta)^{b+n-h-1} \quad \Longrightarrow \quad \theta \mid D \sim \mathrm{Beta}(a+h,\ \ b+n-h)$$

The update is **adding counts**: a and b act like yeses and noes seen before the data. A prior whose posterior stays in the same family is called **conjugate**.

Watching the Posterior Form



Predict before each click: which way does the curve move after a ✗ — and does it get narrower every time?

What the Posterior Gives You

After 3 of 10, with a flat prior: $\theta \mid D \sim \mathrm{Beta}(4, 8)$.

question	answer	from
most plausible p	$0.30 = h/n$	the peak (mode)
best guess on average	0.33	the posterior mean $\frac{a+h}{a+b+n}$
where is p, 95%?	$[0.11,\ 0.61]$	a **credible interval**
after 1,000 users, 300 back	$[0.27,\ 0.33]$	same model, more data

The peak under a flat prior is exactly h/n. **Lecture 03** keeps only this peak and calls it maximum likelihood — the first shortcut of the course.

How Much Does the Prior Matter?

Two analysts: one with a flat prior $\mathrm{Beta}(1, 1)$, one convinced that p is near a half, $\mathrm{Beta}(10, 10)$.

data	flat prior	"near a half" prior
3 of 10	mean 0.33	mean 0.43
300 of 1,000	mean 0.30	mean 0.30

> With little data the prior matters; with enough data the analysts **agree**. A prior is a stated assumption — it can be argued with, and it washes out.

Mini-Exercise

Prior $\mathrm{Beta}(2, 2)$. Of 10 users, **8** come back.

Is the posterior mean above or below 0.8 — and why?

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Below. The posterior is $\mathrm{Beta}(2 + 8,\ 2 + 2) = \mathrm{Beta}(10, 4)$, mean $10/14 \approx 0.71$. The prior counts two yeses and two noes, so it pulls towards 0.5 — and the pull fades as data grows.

Two Ways to Answer "Where Is p?"

FREQUENTIST

p is fixed; the data is random

The confidence interval is a **procedure** that catches p in 95% of samples. About this one interval, no probability statement is allowed.

Needs: nothing but the sampling model.

BAYESIAN

the data is fixed; p is uncertain

Condition on the sample and get a distribution **over** p: "p lies in this interval with probability 0.95" becomes a true sentence.

Needs: a **prior** — what we believed about p before the data.

The axioms are the same; the question is not. The rest of the course mostly takes the second road — and Lecture 03 starts with its cheapest version: keep only the peak.

Takeaway

An unknown parameter is just another unknown: give it a **prior**, condition on the data, and the **posterior** says what the data justifies — as a peak, a mean and an interval. For yes/no data with a Beta prior the update is **adding counts**, and the prior's influence fades as data grows.

Image Credits & References

Sheldon, Ross and others (2002). A first course in probability. *Pearson Education India*.